

Smart Educational Companion: AI-Powered App for Personalized Learning



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Smart Educational Companion: AI-Powered App for Personalized Learning

This Project is Presented to

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In partial fulfillment

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BS in Software Engineering

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This final year project is submitted to the name of Department of Computer Science as partial fulfillment of the requirement for the award of degree of BS Software Engineering.

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*Dedicated to our parents,
who have always been our biggest supporters
and a constant source of inspiration*

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Project Summary

The smart educational Companion is an AI-powered mobile application outlined to improve the learning encounter for understudies in Pakistan. This imaginative stage offers a all encompassing approach to instruction by joining seven key modules that address major torment focuses confronted by understudies and instructors alike. These modules are the Educational programs Module, Timetable Module, and Login Module, Notification Module, Test Module, Chatbot Module, Achievements Module.

The application utilizes cutting-edge innovations like Respond Local for cross-platform improvement, permitting it to run on both Android and iOS systems (right now will make it for Android only). It too highlights a strong backend fueled by Python and employments MongoDB for effective information capacity and recovery, guaranteeing that the app capacities smoothly.

The essential objective of this framework is to give a personalized learning involvement for understudies. It does this by mechanizing key instructive forms, such as course timetable administration, test era, and educational modules following. The AI-powered chatbot encourage upgrades the system's usefulness by advertising real-time clarifications and answers to understudy inquiries, all based on the educational modules substance. Questions are just based on text to text. Moreover, the framework makes a difference instructors by computerizing regulatory errands and giving profitable experiences into understudy progress.

By bridging the hole between conventional instruction and present day innovation, the Keen Instructive Companion points to make learning more open, intuitively, and compelling, especially in districts where instructive assets are constrained. The framework will to begin with center on Course 9 (Material science, Chemistry, Computer Science) subjects, with plans to extend to other subjects and incorporate more progressed highlights in future forms. This extend is a step toward making a advanced classroom environment that not as it were underpins learning but moreover expects and reacts to understudy needs powerfully.

TABLE OF CONTENTS

1	Introduction	xv
1.1	Introduction	1
1.2	Significance of the Developed Solution	2
1.3	Problem Statement	3
1.4	Scope and Objectives	3
1.5	System Limitations/Constraints	5
1.6	Project Deliverables	5
1.7	Relevance to Course Modules	6
1.8	Chapter Summary	7
2	Literature Review	9
2.1	Existing Educational Applications	10
2.2	Existing AI-Powered Learning Systems	11
2.3	Curriculum-Aligned E-Learning Solutions	11
2.4	Related System Analysis	12
3	Requirement Analysis	14
3.1	User Classes and their Key Characteristics	15
3.2	Requirement Identifying Techniques	16
3.3	Functional Requirements	16
3.3.1	FR-1: User Authentication and Role-Based Access	17
3.3.2	FR-2: Student Dashboard	17
3.3.3	FR-3: Learning Content Delivery	19
3.3.4	FR-4: Quiz Management	19
3.3.5	FR-5: AI Chatbot for Academic Support	20
3.3.6	FR-6: Student Progress Tracking	20
3.3.7	FR-7: Teacher Dashboard	21
3.3.8	FR-8: Super Administrator Controls	21
3.3.9	FR-9: Notifications and Alerts	22

3.4	Non-Functional Requirements	22
3.4.1	Reliability	22
3.4.2	Usability	23
3.4.3	Performance	23
3.4.4	Security	24
3.4.5	Performance	24
3.4.6	Security	25
3.5	External Interface Requirements	25
3.5.1	User Interface Requirements	25
3.5.2	Software Interfaces	26
3.5.3	Hardware Interfaces	27
3.5.4	Communication Interface Requirements	27
3.6	Chapter Summary	28
4	Design and Architecture	29
4.1	Modules	30
4.1.1	Curriculum Module	30
4.1.2	Notification Module	30
4.1.3	Quiz Module	30
4.1.4	Login Module	31
4.1.5	Chatbot Module	31
4.1.6	Timetable and Attendance Module	31
4.1.7	Achievement Module	31
4.2	Architectural Design	32
4.3	Design Models	33
4.3.1	Activity Diagrams	33
4.3.2	Use Case Diagram	37
4.3.3	Sequence Diagrams	39
4.3.4	State Transition Diagrams	43
4.4	Data Design	47
4.4.1	Data Dictionary	47

4.5	Human Interface Design	52
4.5.1	Screen images	53
4.5.2	Screen Objects and Actions	56
4.6	Algorithms	57
4.7	Tools and Technologies	61
4.8	User Interface	63
4.9	Chapter Summary	66
5	Testing and Evaluation	67
5.1	Chapter Overview	68
5.2	Testing Introduction	68
5.3	Unit Testing	68
5.3.1	Unit Test Cases	68
5.4	Functional Testing	70
5.4.1	Functional Test Cases	70
5.5	Integration Testing	72
5.5.1	Test Cases for Integration Testing	72
5.6	Chapter Summary	74
6	Conclusion and Future Work	75
6.1	Chapter Overview	76
6.2	Conclusion	76
6.3	Future Work	77
6.4	Summary of Chapter	79
7	References	80

LIST OF FIGURES

Fig 4.1	Proposed system architecture	33
Fig 4.2	Teacher Activity Diagram	35
Fig 4.3	Super Administrator Activity Diagram	36
Fig 4.4	Student Activity Diagram	37
Fig 4.5	Use Case Diagram for Student, Teacher, and Super Administrator	38
Fig 4.6	Sequence Diagram for Teacher	40
Fig 4.7	Sequence Diagram for Super Administrator	41
Fig 4.8	Sequence Diagram for Student	42
Fig 4.9	Teacher State Transition Diagram	43
Fig 4.10	Super Administrator State Transition Diagram	44
Fig 4.11	Student State Transition Diagram	45
Fig 4.12	System Class Diagram	46
Fig 4.13	Login Screen	54
Fig 4.14	Role Selection Screen	54
Fig 4.15	Student Dashboard	54
Fig 4.16	Timetable Screen	55
Fig 4.17	Curriculum Screen	55
Fig 4.18	Quiz Screen	55
Fig 4.19	Login Screen	63
Fig 4.20	Role Selection Screen	63
Fig 4.21	Student Dashboard	63
Fig 4.22	Curriculum Screen	63
Fig 4.23	Quiz Screen	64
Fig 4.24	Attendance Screen	64
Fig 4.25	Chatbot Screen	64
Fig 4.26	Achievements Screen	64
Fig 4.27	Notification Screen	65

Fig 4.28	Leader Board Screen	65
Fig 4.29	Students Management Screen	65
Fig 4.30	Upload Schedule Screen	65

LIST OF TABLES

Table 2.1	Relevant System Analysis with the Developed Solution	12
Table 3.1	User Classes and Their Characteristics	15
Table 3.2	User Authentication and Role-Based Access	17
Table 3.3	Student Dashboard	18
Table 3.4	Learning Content Delivery	19
Table 3.5	Quiz Management	19
Table 3.6	AI Chatbot for Academic Support	20
Table 3.7	Student Progress Tracking	20
Table 3.8	Teacher Dashboard	21
Table 3.9	Super Administrator Controls	21
Table 3.10	Notifications and Alerts	22
Table 4.1	Data Dictionary for Smart Educational Companion	48
Table 4.2	Technical Stack with Implementation Rationale	61
Table 5.1	Unit Test Cases	69
Table 5.2	Functional Test Cases	71
Table 5.3	Integration Test Cases	73

Chapter 1

Introduction

Chapter Overview

This chapter provides an overview of the project by presenting its background, underlying motivation, and the core problem being addressed. It describes the overall vision and defines the scope to clarify the direction of the work. In addition, it emphasizes the significance of the project and clearly states the main objectives, along with a brief outline of how the document is organized.

1.1 Introduction

The platform faces its own array of persistent problems, which has been around for quite a long time already. Teachers waste considerable amounts of time on tedious administrative procedures of recording attendance, sending messages and plans, and answering identical questions from students all the time. Meanwhile, students themselves struggle during silent moments between classes when they lack any means to obtain timely assistance for their curricula-related queries. What follows is a system in which everyone tries their best but fails to achieve anything.

In order to resolve the situation, the Smart Instructive Partner was specifically designed. This is a mobile software program combining cutting-edge academic assistance, automation of administrative procedures and monitoring progress all in one tool suited for the Pakistani environment. Primarily, the app is focused on students enrolled at Punjab Board following Lesson 9 and concentrated mainly on courses in Computer Science, Physics and Chemistry since these subjects tend to give the most problems.

Among the services offered, there is instant access to correct answers from an AI-powered assistance, individualized preparation to tests based on personal weaknesses, automated update of schedules, and a motivational system aimed to guarantee constant interest. Every feature introduced into the system has its distinct function, carefully selected due to particular needs arising within the existing educational process by Course 9 learners.

The whole structure is split among three different client segments, each having its own specific function. Undergraduates utilize the platform to prepare for classes independently, while teachers are responsible for carrying out the technical part of uploading attendance data, maintaining materials, organizing and evaluating tests, and updating schedules. Fi-

nally, Super Administrator takes place at a higher hierarchical level, taking care of account additions and removals and ensuring smooth functioning of the platform. Segmentation makes sure that all roles are assigned correctly and every user operates within his or her limitations.

Obviously, such an initiative can be regarded as extremely helpful to any teacher right off the bat. Indeed, he or she will be relieved from many repetitive administrative chores preventing him or her from doing his or her job properly. Nevertheless, this was not the main goal behind creation of the Smart Instructive Partner. Rather, this was intended to provide some additional assistance to educators who had no means to solve these peripheral issues before. Overall, this system can be viewed as an attempt to leverage modern technologies in favor of Pakistan's education sector.

1.2 Significance of the Developed Solution

For Enabling them to benefit from a proper management of timetables, instant push notifications, syllabus-based academic chatbot assistance, adaptive quizzes for personal weaknesses, gaming features for motivation, and super administrator functionality

Who are plagued by several issues such as outdated learning material, poorly scheduled classes, too much burden of administration placed upon teachers, and a total lack of personalized academic support to help students

The Smart Educational Companion

Is Is a mobile application that incorporates AI-driven learning modules along with a well-managed role-based structure for both students, teachers, and super administrators

That Providing them with an intuitive timetable management module, instant push notifications, curriculum-aligned chatbot for syllabus-related academic assistance, adaptive quizzes focused on personal weaknesses, gaming elements to maintain student motivation, and even a super administrator layer for controlling access and managing accounts

Unlike traditional manual school management systems, generic video tutorial websites, or international AI educational tools that have nothing to do with the syllabus at Punjab Board schools in Pakistan

Our product Our product is designed exclusively for use at Pakistan's secondary school level. Students will get a highly structured and interactive academic companion. Teachers

will receive a set of tools that reduce the amount of time spent on administrative tasks. Super administrators will have full control over the management of the platform.

1.3 Problem Statement

It has been estimated that over 90% of schools and colleges located in Pakistan are still operating on entirely manual systems when it comes to the conduct of educational processes. Students will have to deal with constant difficulties in accessing learning resources; they will have to learn from outdated textbooks and there will be no easy way of verifying if information they use matches with the current curriculum. Whenever changes take place, students simply cannot get to know about it promptly. The issue with class scheduling poses an additional challenge. There is no centralized system that would allow distributing changes and updates to the timetable to be made instantly and accurately. As a result, students and teachers often end up missing out on classes because of their schedule. As far as the process of learning itself is concerned, everything will remain unchanged – there are no personalized features of any kind, meaning that a student’s level of knowledge won’t influence how lessons will be conducted. In addition to this, there will be no diagnostic features to detect weaknesses; students falling behind will not have the option of going through their lessons outside of class as well. As for a teacher, he or she will spend too much time on performing non-educational activities, such as managing class schedules, creating quizzes, providing answers to the same questions of his or her students, as well as monitoring the process of attendance. All of this reduces the overall quality of education provided. There will also be no way of controlling access rights to the platform in accordance with the current roster of teachers in the school. In order for everything to work properly, some management features should be incorporated in the system; the presence of a super administrator account will prove crucial in this case.

1.4 Scope and Objectives

Smart Educational Companion is an AI-driven educational application targeted at Class 9 students in accordance with the Punjab Board curriculum, starting with the introduction of Physics, Chemistry, and Computer Science modules in its first version. Three distinct user types are involved within the project’s architecture: students, teachers, and the super admin.

The students' functionality allows users to log into the system, access their class timetable, take personalized quizzes, see their achievements and interact with an AI chatbot tailored specifically for their academic discipline. The teachers are responsible for operating the application, performing actions such as timetable upload, notifications push, quiz creation, course materials upload, and student progress tracking. Lastly, the super administrator role entails the control over all users in the system, which translates to addition/removal of teachers' accounts as needed.

The underlying technology used in the development of the application includes RAG-based AI chatbots, capable of delivering answers specific to local curriculums, rather than generic AI tutors designed for international students. Another element implemented in the platform is gamification, specifically achievements and badges, helping users gauge their current position in studies and maintain high engagement.

The system will undergo its initial deployment in the first phase of the project is limited to Android systems via React Native technologies, providing maximum accessibility for students and teachers using different mobile devices in Pakistan. Moreover, any out-of-scope functionalities, such as fee management, video lectures or links to online learning platforms are planned not to be included in the initial release. Nevertheless, scalability is a priority in the system's architecture, meaning that expansion in any direction, be it subject addition, other curriculums incorporation, or media format change is possible.

The objectives of the Smart Educational Companion are as follows:

1. Designing an AI-based mobile app based on the Class 9 Punjab Board curriculum for Physics, Chemistry, and Computer Science.
2. Facilitating automated timetable planning and sending real-time notifications about schedules for students and teachers while preventing confusion from manual communication.
3. Designing an AI chatbot based on RAG according to the curriculum to assist the students with their studies.
4. Designing a quiz system to find weak points of each student individually and give prac-

tice tests immediately after that.

5. Including gamification features such as badges to motivate the users and keep them engaged.
6. Offloading the teacher's work on managing quizzes, creating timetables, attendance, and monitoring the performance of students.
7. Assigning a super administrator who can include teachers' accounts on the app or remove them.

1.5 System Limitations/Constraints

LI-1 The system doesn't support any kind of voice query. All queries are done via typing only.

LI-2 Curriculum mapping is available for only the below mentioned topics of class nine under Punjab Board curriculum: Physics, Chemistry, Computer Science

LI-3 Quiz creation supports objective type questions only. The following question types do not fall within current capabilities:

Written subjective questions

Diagram-based questions

LI-4 The system does not support any image processing tasks. This includes inability to:

Upload images, either for analysis purposes or otherwise

Submit images or analyse them

LI-5 The role of super administrator is limited to management of accounts only. The super admin has permissions to create and delete accounts of teachers, but cannot access any student data.

1.6 Project Deliverables

- Mobile application built in React Native and using the Expo framework with an Android operating system.

- AI Chatbot created using Retrieval-Augmented Generation (RAG) which is tailored to the syllabus of Punjab Board Class 9 including Physics, Chemistry, and Computer Science subjects.
- Quiz module with an adaptive approach along with automated evaluation and personalized feedback for individual users.
- Timetable management module where teachers can post timetables and send notification to students.
- Game and achievement tracking module to measure progress of students and engage them over a prolonged period of time.
- Teacher's dashboard to handle course content, quizzes, attendance, and student notifications.
- Super-administrator panel that allows to add/remove teacher accounts from the system.
- Attendance module with a feature to upload attendance records by teachers and view their individual record by students.
- Complete documentation for the project containing SRS, System Design Document, and the Final Report.
- Final presentation and demonstration of the working solution.

1.7 Relevance to Course Modules

The **Smart Educational Companion** The Smart Educational Companion project is built upon the concepts and skills acquired during the Bachelor of Software Engineering (BSE) program. The following courses have played a significant role in guiding the design and implementation of the system, contributing in various ways as outlined below:

- **Software Requirement Engineering (SRE):** Leveraged extensively during the requirement gathering process to extract, analyze, and verify functional and non-functional

requirements for all user roles, namely the student, teacher, and the super administrator.

- **Mobile Application Development (MAD):** Contributed to building the application itself using React Native and Expo in terms of implementing navigation flow, designing components and managing different screens depending on role of each user.
- **Machine Learning:** Contributes to the adaptive quiz engine and chatbot functionalities by leveraging machine learning algorithms to identify weak areas of the student and providing tailored quizzes for practice.
- **Software Design and Architecture:** Provided guidance on the three tier role hierarchy and modularity of the application in addition to decisions related to architecture and integration of frontend with backend in python.
- **Software Quality Engineering (SQE):** Applied in the formulation of testing and quality assurance processes through the process of verification and validation performed during development of all modules.
- **Data Structures and Algorithms (DSA):** Used to facilitate retrieval of content from the curriculum within the RAG chatbot in addition to quiz generation and grading mechanism.
- **Operating Systems (OS):** Principles of process and resource management provide guidance for understanding the behavior of the application across multiple devices in Pakistan.
- **Software Project Management (SPM):** Leveraged to organize work and tasks assigned to individual members throughout the development process.

1.8 Chapter Summary

This chapter provides the foundation of the whole document. The introduction to the Smart Educational Companion begins with the introduction to the current state of education in Pakistan, with such problems as outdated resources, poor scheduling, and the complete

lack of personalized assistance for the students. Section 1.2 highlights the importance of the project, describing it as an opportunity to create a useful localized educational resource developed with a focus on student requirements, the educators, and school administrators. Section 1.3 presents the problem statement, providing more details on the inefficiencies that arise from the manual system used, and also highlighting the problems faced by the teachers and the school administration. In section 1.4, the scope and objectives of the project are clarified. Here, we outline the main tasks performed by the software, who the audience is among the three groups using the application, and what we will not address in this particular version. In section 1.5, the limitations of the project are presented, including such important details as restricted subjects, inability to send multimedia messages through the chatbot, and restricted access to the super administrator account. The deliverables of the project are outlined in section 1.6, including the development of the mobile application itself, creation of the teacher panel and super administrator panel, development of the chatbot, creation of the quiz and attendance modules, and all documentation. Finally, in section 1.7, the project is mapped to the BSE curriculum, highlighting how each subject such as Machine Learning, Mobile Application Development, or Software Project Management played its role.

Chapter 2

Literature Review

Chapter Overview

In this chapter, we will review some of the existing examples of technology-based learning in order to understand what methods have been tested, what has proven successful, and where there are gaps in their application. In other words, this chapter outlines the shortcomings of current approaches to digital tutoring and mobile learning that will be addressed in the creation of the Smart Educational Companion.

2.1 Existing Educational Applications

There is an abundance of platforms like Coursera, Khan Academy, and Duolingo that have revolutionized access to education worldwide. Each of them offers structured learning pathways, internal quiz systems, educational videos, and gamified tracking tools to monitor student progress. Their global outreach and effectiveness cannot be denied. Nevertheless, these learning solutions were designed for different audiences, not Pakistani secondary school students in particular. Herein lies the issue, as it translates into several problems:

- **Misalignment With School Curriculum:** These applications are based on syllabuses developed elsewhere. For instance, a student from Class 9 who needs to prepare for Punjab Board exams will face topics, vocabulary, and questions in these tutorials that are incompatible with the ones in their test papers. Therefore, it would take a considerable amount of extra time to reconcile both sources and complete their task successfully.
- **Lack Of Local Adaptation:** None of these applications take into account the specifics of the Pakistani educational system. That includes the academic calendar, examination format, etc. Therefore, none of them can provide adequate support for a student from Pakistan.

These platforms serve millions of users globally, but for a Class 9 student in Punjab, they remain a poor fit for day-to-day academic preparation.

2.2 Existing AI-Powered Learning Systems

The integration of artificial intelligence into education has produced a new generation of learning tools capable of adapting to individual students, generating practice material automatically, and providing conversational academic support. Studies across multiple contexts have shown measurable improvements in student engagement and retention when AI-driven personalisation is introduced into the learning process.

Despite these results, the same systems carry practical limitations that make them difficult to apply in a local Pakistani school setting:

- **High Infrastructure Cost**
- **No Syllabus-Specific Training**
- **Missing School-Level Features**
- **Overly Generic Output**

These gaps point directly to the need for a system that brings AI capability into a locally grounded, curriculum-specific context rather than importing a general-purpose tool and expecting it to fit.

2.3 Curriculum-Aligned E-Learning Solutions

The Punjab Government and various private publishers have made efforts to bring education online through digital portals, downloadable PDF textbooks, and recorded video lectures. These resources were particularly relied upon during the disruptions to in-person schooling in recent years and remain widely referenced by students looking for additional material. Their limitations, however, are significant enough to prevent them from functioning as complete learning companions: Static and Non-Adaptive Content, No Real-Time Interaction, No Engagement Mechanisms, No AI or Personalisation Layer. These portals do not generate quizzes, identify weak areas, or adapt in any way to how an individual student is progressing through the material.

These materials play the role of references, but they do not come close to what is required for adequate preparation.

2.4 Related System Analysis

Table 2.1 below demonstrates some examples of current approaches that differ from those of the Smart Educational Companion, pointing out their weaknesses as compared to the latter one.

Table 2.1 Relevant System Analysis with the Developed Solution

Application	Limitations	How This Project Addresses It
Khan Academy	Follows international syllabi with no alignment to Punjab Board content; primarily English-medium with no localised examples relevant to Pakistani students.	Chatbot trained directly on Punjab and Rawalpindi Board textbooks, with responses grounded in the exact curriculum local students are examined on.
Punjab e-Learn Portal	Delivers static PDFs and pre-recorded lectures with no AI support, no real-time question answering, and nothing to drive ongoing student engagement.	AI-powered chatbot with real-time Q&A, adaptive quiz generation, and gamified progress tracking built into the same platform.
YouTube Educational Channels	Unstructured content with inconsistent quality, frequent distractions, and no connection to timetable management, notifications, or academic tracking.	Structured, syllabus-aligned content embedded within a platform that also handles timetable updates, notifications, and attendance in one place.
Generic AI Tutors (e.g., ChatGPT)	Not trained on Pakistan's curriculum; no quiz integration; no attendance, timetable, or institutional management features; no role-based access control.	Localised AI chatbot aligned with Punjab Board syllabus, integrated with quizzes, timetable, attendance, and a three-role access structure covering students, teachers, and a super administrator.

Chapter Summary

In this chapter, I conducted an extensive review of the current technologies used in educational settings in order to determine their areas of success and shortcomings for the needs of Pakistani secondary school students. Coursera, Khan Academy, and Duolingo are examples of successful learning platforms that offer comprehensive courses but are targeted at global learners. These educational platforms do not provide curriculum alignment or even academic contexts that could be useful for students enrolled in the Punjab Board syllabus. Artificial intelligence (AI) learning platforms are another example of advanced technological products with great potential to provide personalised learning courses. However, these technologies require significant funding, and they also lack the necessary information about the Punjab Board curriculum content and school-level requirements, such as planning lessons or sending notifications. In addition, there are various e-learning resources developed by both government and non-governmental publishers in Pakistan that provide curriculum-aligned courses. However, the content provided by these publishers is static; students use these resources as electronic versions of traditional textbooks with no additional features.

The table comparing the current technological solutions and their relevance for the proposed Smart Educational Companion revealed that none of the reviewed technologies provide a combination of local curriculum content, AI-based personalised learning, and school-level management tools, including planning, notifying, and controlling students' activities.

Chapter 3

Requirement Analysis

Chapter Overview

In this chapter, information on how requirements for the Smart Educational Companion were elicited, defined, and verified is provided. User classes who will use the system are identified; elicitation methods utilized to elicit requirements; and functional and non-functional requirements are defined and illustrated. This helps ensure that all functionalities included in the system relate to a user need or system requirement.

3.1 User Classes and their Key Characteristics

There are three types of users for whom the system is being developed. Each of the users has specific roles, accesses to be allowed, and requirements of the system. Table 3.1 gives an overview of the user classes.

Table 3.1 User Classes and Their Characteristics

User Class	Characteristics
Students	The users of the platform. Class 9 students use the platform to learn curriculum-based concepts in Physics, Chemistry, and Computer Science, try out adaptive quizzes, converse with the chatbot about their studies, see their schedule and attendance status, and monitor their progress. Students
Teachers	Operational users responsible for managing the academic content and day-to-day activities on the platform. They will upload timetables and attendance details, conduct quizzes, send push notifications to their students, and check the performance of the students. They need a good dashboard that provides a view of the activities of the students without being complicated.
Super Administrator	Super Administrator is responsible for creation and deletion of teacher profiles, making sure that only authorized people can access the teacher panel. This position does not deal with any student data directly; its main function lies in ensuring the organized operation of the entire program.

3.2 Requirement Identifying Techniques

The collection of requirements for a system that caters to three distinct types of users was achieved through several means since a single method may not have brought out all the requirements needed. Three methods were employed to ensure completeness of requirements gathered.

- **Use Case Modelling:** The use cases were developed to describe how the system is used by three distinct user classes: the students, the teachers, and the super administrators. This helped describe the expected outcome of each use of the system and was used to limit the scope of each feature to a single user.
- **Storyboarding and Prototyping:** Considering that the system was an app whose functionality was dependent on the user interface, storyboards and a front-end prototype was developed before coding of the system began. This enabled the identification of issues relating to the usability of the system before starting the coding process.
- **Interviews and Questionnaires:** Informal interviews were held with students to find out the specific inconveniences they faced using the manual system such as missing schedules, need for academic assistance at a moments' notice, and inability to track progress. In conjunction with interviews, questionnaires were used for the same purpose.

3.3 Functional Requirements

This section presents and explains the functional requirements that define the behavior and capabilities of the Smart Educational Companion. Each requirement specifies something the system must actively do to serve its users students, teachers, or the super administrator. Requirements are expressed in a structured format with a unique identifier, title, description, source, rationale, dependencies, and priority level.

3.3.1 FR-1: User Authentication and Role-Based Access

Table 3.2 is reflecting the very first functional requirement which is showing the user authentication and Role Based Access.

Table 3.2 User Authentication and Role-Based Access

Identifier	FR-1
Title	User Authentication and Role-Based Access
Requirement	The system shall allow all user classes students, teachers, and the super administrator to log in securely using verified credentials and access only the features and data assigned to their respective role. A student login shall not expose teacher or super administrator functionality, and vice versa.
Source	System Requirement
Rationale	The system implements role-based access control to enforce permissions, ensuring that each user can access only the modules relevant to their role.
Dependencies	None
Priority	High

3.3.2 FR-2: Student Dashboard

Table 3.3 is reflecting the second functional requirement for student dashboard.

Table 3.3 Student Dashboard

Identifier	FR-2
Title	Student Dashboard
Requirement	The system shall provide every student with a customized dashboard tailored to their needs that serves as their central access point for curriculum-aligned learning content, quizzes, timetable information, attendance records, notifications, and achievement tracking. The dashboard shall reflect data specific to the logged-in student.
Source	Requirement Document
Rationale	Consolidating all student-facing features within a single dashboard reduces navigation complexity
Dependencies	FR-1
Priority	High

3.3.3 FR-3: Learning Content Delivery

Table 3.4 is reflecting the third functional requirement which is Learning Content Delivery.

Table 3.4 Learning Content Delivery

Identifier	FR-3
Title	Learning Content Delivery
Requirement	The system shall deliver subject-specific course materials for Physics, Chemistry, and Computer Science in alignment with the Class 9 Punjab Board syllabus. Content shall be uploaded by teachers through the teacher panel and made accessible to students through their dashboard.
Source	Punjab Board Academic Standards
Rationale	Students require access to materials that directly correspond to what they are being examined on.
Dependencies	FR-1, FR-2
Priority	High

3.3.4 FR-4: Quiz Management

Table 3.5 is reflecting the forth functional requirement which is quiz management.

Table 3.5 Quiz Management

Identifier	FR-4
Title	Quiz Management
Requirement	The system shall allow teachers to create and publish subject-specific quizzes. Students shall be able to attempt these quizzes through their dashboard and receive results upon submission. Each result shall include a score along with AI-generated explanations for incorrect responses.
Source	Requirement Document
Rationale	Regular assessed practice with immediate feedback is one of the most reliable ways to consolidate learning.
Dependencies	FR-1, FR-3
Priority	High

3.3.5 FR-5: AI Chatbot for Academic Support

Table 3.6 is reflecting the fifth functional requirement which is AI chatbot for academic support

Table 3.6 AI Chatbot for Academic Support

Identifier	FR-5
Title	AI Chatbot for Academic Support
Requirement	The system shall provide an AI-powered chatbot that students can use to ask academic questions related to the Class 9 syllabus across Physics, Chemistry, and Computer Science. The chatbot shall use (RAG) to produce responses.
Source	Project Scope
Rationale	Students frequently encounter concepts they do not understand outside of class hours with no reliable way to get an immediate, curriculum-relevant answer. The chatbot fills this gap
Dependencies	FR-1, FR-3, FR-4
Priority	High

3.3.6 FR-6: Student Progress Tracking

Table 3.7 is reflecting the sixth functional requirement which is Student progress tracking.

Table 3.7 Student Progress Tracking

Identifier	FR-6
Title	Student Progress Tracking
Requirement	The system shall track each student's quiz performance over time and generate achievement summaries that identify subject-level strengths and areas requiring further attention. Progress data shall be visible to the student through their dashboard. Teachers shall also have access to aggregated performance data
Source	Teacher Feedback
Rationale	Making progress data available to both parties ensures that weak areas are identified early and that the support provided is directed where it is actually needed.
Dependencies	FR-1, FR-2, FR-4
Priority	High

3.3.7 FR-7: Teacher Dashboard

Table 3.8 is reflecting the seventh functional requirement which is teachers dashboard.

Table 3.8 Teacher Dashboard

Identifier	FR-7
Title	Teacher Dashboard
Requirement	The system shall provide teachers with a dedicated dashboard through which they can upload and manage timetables, publish course materials, upload attendance records, send notifications to students, and view individual and overall student performance data.
Source	System Requirement
Rationale	Teachers need a single, organised workspace that covers both content management and student oversight.
Dependencies	FR-1, FR-6
Priority	High

3.3.8 FR-8: Super Administrator Controls

Table 3.9 is reflecting the eighth functional requirement which is Super Administrator Controls.

Table 3.9 Super Administrator Controls

Identifier	FR-8
Title	Super Administrator Controls
Requirement	The system shall provide a super administrator panel through which the super admin can create new teacher accounts, remove existing teacher accounts, and view the list of currently registered teachers.
Source	Project Requirement Specification
Rationale	As the platform scales and teacher accounts change over time, there must be a clearly defined role responsible for controlling who has operational access to the system. Keeping this responsibility separate from the teacher role prevents
Dependencies	FR-1
Priority	High

3.3.9 FR-9: Notifications and Alerts

Table 3.10 is reflecting the ninth functional requirement which is Notifications and alerts.

Table 3.10 Notifications and Alerts

Identifier	FR-9
Title	Notifications and Alerts
Requirement	The system shall enable teachers to compose and send notifications to students regarding timetable changes, newly uploaded course materials, quiz announcements, and other academic updates. Notifications shall be delivered in real time and displayed within the student's notification screen inside the application.
Source	Stakeholder Feedback
Rationale	One of the most common sources of confusion in the current manual system is that schedule and content changes do not reach students reliably or on time. A built-in notification mechanism ensures that updates are communicated instantly and that students do not miss information that affects their academic activities.
Dependencies	FR-1, FR-7
Priority	Medium

3.4 Non-Functional Requirements

Non-functional requirements define the standards the Smart Educational Companion must meet beyond what it does covering how reliably it operates, how quickly it responds, how securely it handles data, and how comfortably real users can navigate it. Each requirement below is stated in measurable terms so that it can be tested and verified during and after development.

3.4.1 Reliability

Reliability requirements define the expectations around system uptime, failure frequency, and recovery time. Given that students rely on the platform for time-sensitive activities such as viewing timetables and submitting quizzes, unplanned downtime carries a direct academic cost.

- **REL-1:** The system shall ensure continuous availability with an uptime of at least 99.5% across each academic year, excluding any periods of pre-announced scheduled

maintenance.

- **REL-2:** The Mean Time Between Failures (MTBF) for the system shall be no less than 200 hours of continuous operation.
- **REL-3:** Following any unplanned system failure, critical services including the student dashboard, timetable display, and quiz module shall be fully restored within 2 hours of the failure being identified.

3.4.2 Usability

Usability requirements define how easily different users can learn and operate the system. Because the platform serves students who may have limited experience with mobile applications, and teachers who need to complete administrative tasks quickly, the interface must be intuitive enough to use without training while remaining functional enough to support the full range of features each role requires.

- **USE-1:** A new student shall be able to independently navigate the core features of the application accessing course materials, viewing the timetable, and attempting a quiz within 15 minutes of first use, without requiring any external guidance or documentation.
- **USE-2:** The system shall maintain a consistent interface across all three user roles students, teachers, and the super administrator with clearly labelled navigation elements, recognisable icons, and unambiguous action labels that behave the same way regardless of which screen or module the user is currently in.

3.4.3 Performance

Performance requirements define the speed and capacity the system must sustain under normal and peak usage conditions. These are particularly relevant given that multiple students may access the platform simultaneously during exam preparation periods or when a teacher pushes a timetable update to the entire class.

- **PER-1:** At least 95% of content screens including course material pages, quiz questions, and result summaries shall finish loading within 3 seconds under normal network conditions.

- **PER-2:** The system shall handle a minimum of 200 concurrent active users without any measurable degradation in response time or feature availability.
- **PER-3:** Following a quiz submission, the system shall deliver the student's results along with AI-generated explanations for any incorrect answers within 5 seconds.

3.4.4 Security

Security requirements define how the system protects user data and prevents unauthorised access. Given that the platform holds student performance records and teacher-uploaded academic content, maintaining strict access boundaries between roles is not optional it is a baseline expectation.

- **SEC-1:** The system shall enforce role-based access control at the application level, ensuring that students, teachers, and the super administrator can only reach the features and data assigned to their specific role. No cross-role data access shall be permitted under any circumstances.
- **SEC-2:** All data exchanged between the mobile application and the backend server shall be transmitted over encrypted channels using industry-standard protocols, ensuring that sensitive information cannot be intercepted in transit.
- **SEC-3:** The system shall automatically lock any user account for a period of 15 minutes following five consecutive failed login attempts from the same account, regardless of user role.

3.4.5 Performance

The performance requirements specify the rate and capacity that the system should maintain throughout its operation. This aspect is important in consideration of the fact that several students might concurrently interact with the system while preparing for an exam or when a teacher sends out an updated timetable to the entire group.

- **PER-1:** At least 95% of content screens comprising course materials, quizzes, and quiz results must fully load within three seconds under normal network conditions.

- **PER-2:** The system must support a minimum of 200 simultaneous active users without experiencing any reduction in its responsiveness and accessibility.
- **PER-3:** Once a student submits their quiz, the system must return the test results alongside artificial intelligence-generated explanations of the mistakes within five seconds.

3.4.6 Security

Security requirements determine the system's capabilities in ensuring the confidentiality and integrity of its data, as well as the restrictions it imposes on any unauthorised access. In view of the fact that the application contains student performance records and academic content uploaded by teachers, there must be no compromise on the boundaries of access between users' roles.

- **SEC-1:** Role-based access controls should be applied at the application level to restrict access to particular features and data depending on each role. Cross-role data access must be prevented.
- **SEC-2:** Data exchange between the mobile application and the server must take place over secure connections that comply with industry standards.
- **SEC-3:** Any user account must be locked out of the system for fifteen minutes after five consecutive failed login attempts.

3.5 External Interface Requirements

This section describes the conditions under which the Smart Educational Companion communicates with the users of the application and with other hardware or software components. It guarantees that the communication will take place effectively in any case, regardless of whether this concerns the screens of the application, services used, operating devices, and even network conditions.

3.5.1 User Interface Requirements

The design of the application complies with the modern standards for UI/UX in applications, which prioritize usability for all three roles at once. The design should suit both the

student in the process of studying new material and the super-administrator working with the profiles of teachers without any inconveniences.

- The interface of the application shall comply with the existing rules for UI/UX in modern mobile applications and adhere to Material Design conventions.
- Navigation shall include Home, Back, and Profile options available on all the application screens.
- The typography shall be consistent throughout all the application with appropriate differentiation in fonts used for headings, body text, and labels.
- Icons representing actions within the application shall be consistent across all modules and look identical.
- There shall be consistent usage of colours in accordance with the blue and white theme of the application across all application screens and the three types of user interfaces.
- The layout of messages (errors, system prompts, and others) displayed within the application shall be standardized.
- Layouts of all application screens shall be optimized for standard Android smartphones.

3.5.2 Software Interfaces

The Smart Educational Companion interacts with many internal and external software modules. Each interface defined below needs to be handled correctly by the system for the platform to function accordingly.

- **Database:** The system should connect with a NoSQL database (MongoDB) where data concerning the student's profile, quiz performance, attendance, course material, timetable, and teacher's accounts managed by the super administrator can be retrieved and updated. A vector database shall be kept in order to allow the functionality of the RAG-based chatbot, which stores the curriculum content in vector format.

- **AI Module:** For the purposes of developing the chatbot and quiz explanations, the Retrieval-Augmented Generator model shall be used. Student queries will be embedded by using a Hugging Face embedding model, the relevant context shall be retrieved from a vector database, and finally passed along with it to the open-source language models (Gemma, LLaMA, etc.) in order to generate answers that are specific to the Punjab Board syllabus.
- **Operating System:** The system should be able to run on an Android Oreo 8.0 operating system and its newer versions.

3.5.3 Hardware Interfaces

As a mobile application, the Smart Educational Companion relies solely on the capabilities provided by the smartphone hardware. The following are hardware interfaces requirements:

- The application should operate on devices having 4 GB RAM and at least 16 GB free internal storage.
- All the user input should be made possible via the touch screen and on-screen virtual keyboard without requiring any other form of hardware.
- The application should use any of the Internet connections available to the device (Wi-Fi and cellular network) to connect to the backend server for carrying out operations in real time.
- The system will use the device notification facility to alert students about new timetable uploads and quiz notifications.

3.5.4 Communication Interface Requirements

Below are the communication requirements that define how the system communicates with its users and other services.

- **Push Notifications:** Push notifications must be provided to students whenever there are changes made to the class timetable, when new material is uploaded, and when announcements are posted. These notifications will appear on the notification screen within the app and, if supported by the phone OS, as an in-device notification.

- **Automated Emails:** Automated emails will be sent to registered users to verify accounts and deliver an OTP in case there is a password reset. Both student and teacher accounts will benefit from this.

3.6 Chapter Summary

This chapter detailed all the requirements of the Smart Educational Companion regarding its three roles of users; students, teachers, and the super administrator.

Starting with the identification of these three roles of users and what their responsibilities are, along with their specific needs from the system. After which, it described the methods used to identify those requirements, namely, using use case modelling to determine role specific interactions, prototyping to make sure navigation and layouts were valid prior to development, as well as using interviews and questionnaires.

Finally, it covered the functional requirements detailing nine specific requirements regarding student and teacher dashboards, curriculum related learning material delivery, quiz management, the RAG based chatbot, tracking student progress, managing super administrator accounts, and the notification system. The non-functional requirements addressed reliability, usability, performance, and security each with measurable targets. Of particular note is the security requirement for strict role-based access control, which ensures that students, teachers, and the super administrator can only ever reach the parts of the system that belong to their role.

Finally, the external interface requirements covered the user interface standards applied consistently across all three role-based panels, the software interfaces connecting the application to MongoDB, the vector database, and the RAG pipeline, the hardware constraints for the Android devices the application runs on, and the communication interfaces handling push notifications and email-based account management.

Taken together, this chapter provides a complete and structured account of what the Smart Educational Companion must do, for whom, and under what conditions forming the specification that the system design in the following chapter is built upon.

Chapter 4

Design and Architecture

Chapter Overview

This chapter covers the design decisions made for the Smart Educational Companion and the reasoning behind them. It begins by describing the individual modules that make up the system, then presents the overall architectural approach, followed by the diagrams that capture the system's structure and behaviour including use case, activity, class, sequence, and state transition diagrams. Furthermore, the chapter highlights the aspect of data design and describes how various modules are designed and linked to each other.

4.1 Modules

The SECS system has well-defined modules which perform particular functions. Such approach makes it easy to expand the system and/or make changes without affecting other components. Each role that the software targets students, teachers, and the super administrator can access only those modules relevant to their role in the system.

4.1.1 Curriculum Module

This module provides access to the courseware in the areas of Physics, Chemistry, and Computer Science, according to the Punjab Board schedule. Content for all topics is divided by subjects. Therefore, students will be able to look at what has been taught for each week. They will also be able to use the chatbot interface to get more information about particular curricular topics.

4.1.2 Notification Module

This module deals with all messages from teachers to students. Teachers write announcements which concern timetable alterations, updates of any kind, new materials available on the website, quizzes deadline, etc., from the teacher's dashboard. These notifications appear instantly on the students' notification page or even as native push notifications if possible.

4.1.3 Quiz Module

This module performs all activities related to the quiz cycle on this software. All tests based on the current syllabus can be created by the teacher via his/her dashboard. After attempting quizzes, students will get immediate results that include feedback for each incorrect

answer and detailed explanation provided by the AI algorithm. This module is linked to the achievements system that offers badge awards and progression based on quiz scores.

4.1.4 Login Module

This module is responsible for authentication and role-based access control for all users. For students and teachers, access is granted using their registered email and personal password. The password of the super administrator account should be entered with one-time passcode authentication as an extra level of security. On successful validation, every user is sent to the corresponding user role interface: student dashboard, teacher panel or super administrator panel without any cross-role permissions.

4.1.5 Chatbot Module

Chatbot module helps students seek educational assistance in a conversational manner. Student can ask queries on any topic related to Class 9 Physics, Chemistry or Computer Science, which will be answered on the basis of Punjab Board curriculum content via the RAG pipeline. Furthermore, explanations get triggered automatically by the system whenever a student gets a question wrong in quizzes without needing to raise another question.

4.1.6 Timetable and Attendance Module

The timetable and attendance module controls the scheduling and record keeping for both the timetables and attendance of all the users registered on the system. Teachers upload timetables and attendance details on their dashboard with OCR functionality through Tesseract API helping to extract structured data from the uploaded documents. In addition, students can view their current timetables, upcoming schedules, and monitor their attendance details while super administrators do not have anything to do with this module.

4.1.7 Achievement Module

Finally, the achievement module introduces gamification aspect for students to collect badges and points based on their quiz scores and number of times they participate. Thus, providing them with quantifiable metric of their learning journey. Also, teachers can analyze summary of achievements of their students and know who are consistent and needs some more attention.

4.2 Architectural Design

Smart Educational Companion follows a **client-server architecture** consisting of multiple tiers where the UI, business logic and data management are separated in different layers. Separation of concerns makes the system maintainable, easily testable individually and scalable to accommodate more changes without affecting the rest of the code base.

Overall, the architecture is divided into three major layers

- **Presentation Layer:** It includes mobile application interfaces for all the three user roles namely the student dashboard, teacher panel and super administrator panel. It takes care of everything related to what user views or interacts such as navigation among screens, user inputs and display of information returned from the application layer.
- **Application Layer:** Contains the core business logic of the system. This includes the RAG-based chatbot engine, the quiz management and evaluation logic, timetable and attendance processing, notification dispatch, achievement tracking, and the role-based access control that determines what each user type can see and do. All API endpoints that the mobile application communicates with are defined and handled at this layer.
- **Data Layer:** Manages all persistent storage for the system, including student records, quiz results, attendance data, course materials, timetable entries, teacher accounts, and the vector database used by the RAG pipeline for curriculum content retrieval. This layer enforces secure and structured access to the underlying MongoDB instance and vector store.

Each layer communicates only with the layer directly adjacent to it the presentation layer calls the application layer through REST API endpoints, and the application layer reads from and writes to the data layer as needed. This keeps the system loosely coupled, meaning a change in one layer does not force changes in the others.

The overall structure of the architecture is illustrated in Figure 4.1.

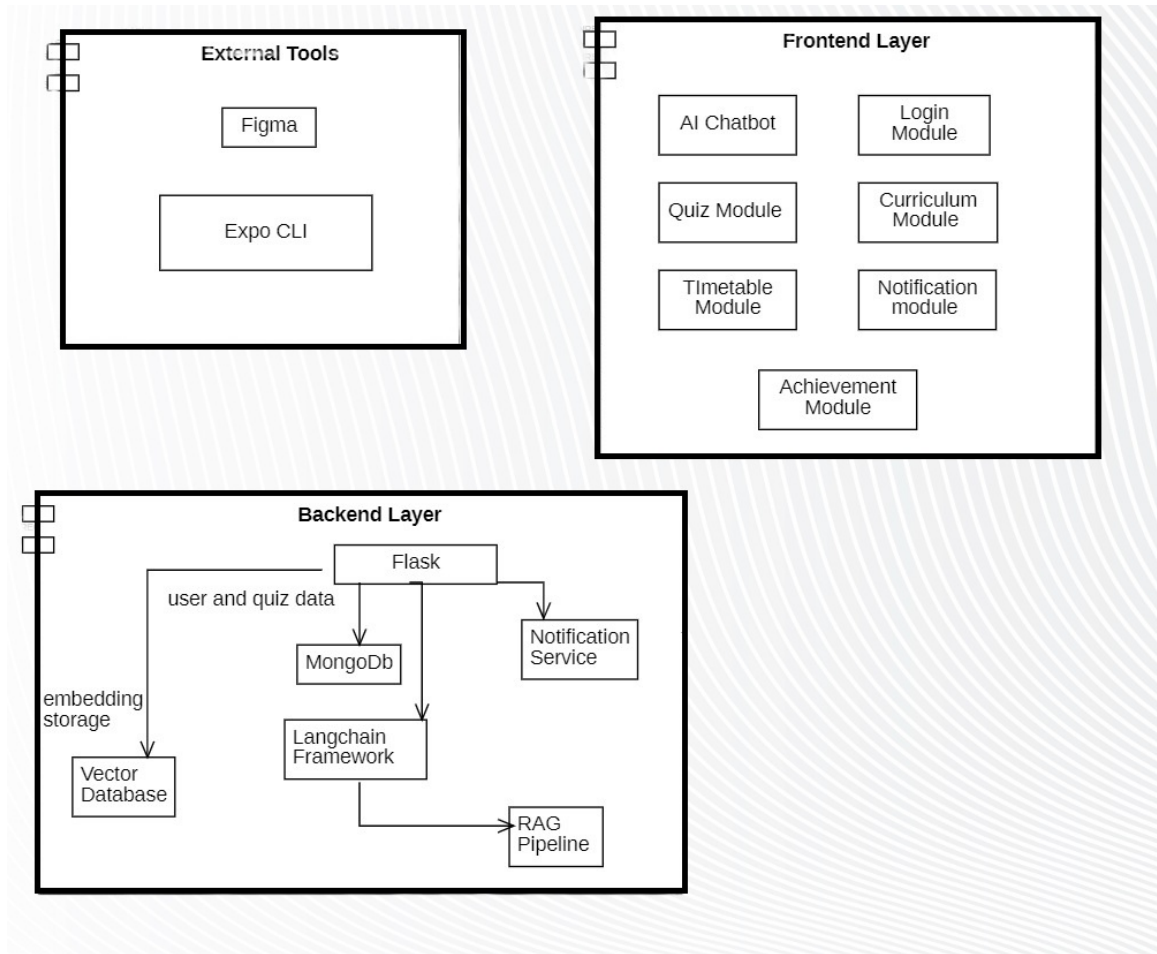


Figure 4.1 Proposed system architecture

4.3 Design Models

4.3.1 Activity Diagrams

Activity diagrams capture the step-by-step workflow of each user role within the Smart Educational Companion. Because the system serves three distinct roles students, teachers, and the super administrator each with different responsibilities and access boundaries, separate diagrams are provided for each role to keep the workflows clear and unambiguous.

Teacher Activity Diagram

The Teacher Activity Diagram (Figure 4.2) illustrates the entire sequence of operations teachers can execute from login to logout. The main actions include:

- **Student Record Management:** Teachers are able to create new accounts for their students or edit existing student information when necessary.

- **Course Materials Management:** Teachers can upload textbooks, notes, and other course material in addition to updating existing materials to stay consistent with the curriculum.
- **Quizzes Management:** Teachers can design new quizzes and assign them to certain topics or subject areas and evaluate their submission outcomes afterward.
- **Attendance Management:** Teachers upload students' attendance information which is then displayed on individual dashboards for students.
- **Notifications Sending:** Teachers are able to type notifications which contain information such as timetables, updated course materials, upcoming quiz deadlines, and other academic-related data.
- **Student Performance Reviewing:** Teachers can analyze students' performance in quizzes and review achievement statistics related to quizzes.
- **Chatbot Utilizing:** Teachers can use the chatbot feature to get answers to questions regarding timetables or curriculum coverage.

It is worth noting that after login, teachers have the opportunity to switch between managing course material, handling quizzes, and analyzing reports in any order; meanwhile, sending notifications can be done in parallel with the above operations.

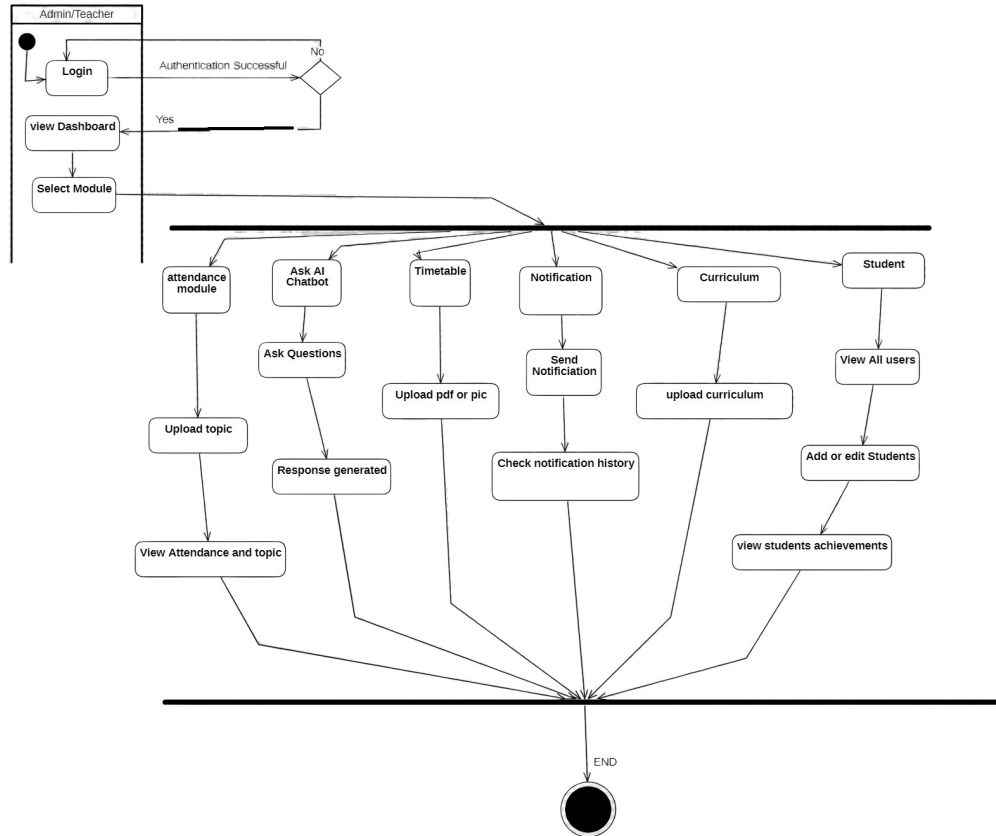


Figure 4.2 Teacher Activity Diagram

Super Administrator Activity Diagram

Workflow of Super Administrator (Super Admin) (see Figure 4.3) is intentionally limited in comparison with the workflow of the teacher since the scope of the super administrator's activities is related to account management at the platform level, not to educational content.

The major activities performed by the super administrator are:

- **Add Teacher Account:** The super administrator can create the teacher's account using the relevant credentials, after which this account will be activated and the teacher will be able to enter the user panel.
- **Remove Teacher Account:** The super administrator can disable or delete the already created account and deprive the particular teacher of the possibility to use the system.
- **List of Registered Teachers:** The super administrator has an option to see the complete list of the teachers whose accounts are currently registered in the system.

OTP verification takes place when entering, and logout completes the workflow. Neither quizzes, educational content, attendance records nor students' data are available in the super administrator's work mode.

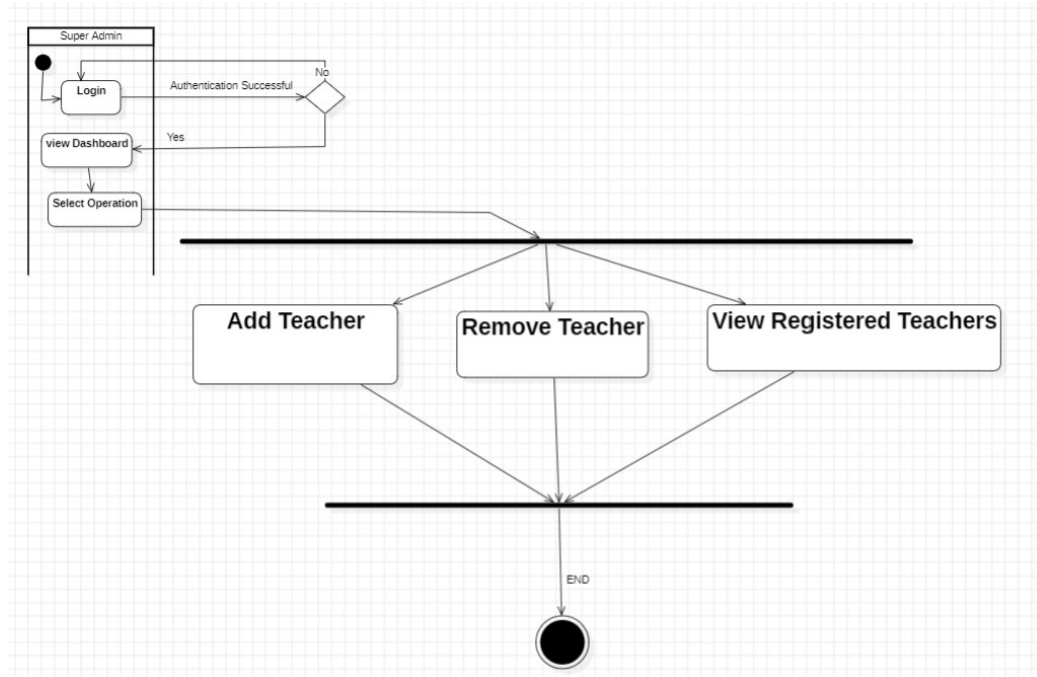


Figure 4.3 Super Administrator Activity Diagram

Student Activity Diagram

Student Activity Diagram (Figure 4.4) depicts the total set of actions that students perform inside the system. The process starts with authentication and terminates at logout; this means that all actions which users perform from login to logout are presented in the diagram. Viewing the material of courses, performing quizzes, reviewing the results, browsing the timetable, monitoring attendance, looking through achievements, and viewing messages from teachers make up the main part of the student activity list.

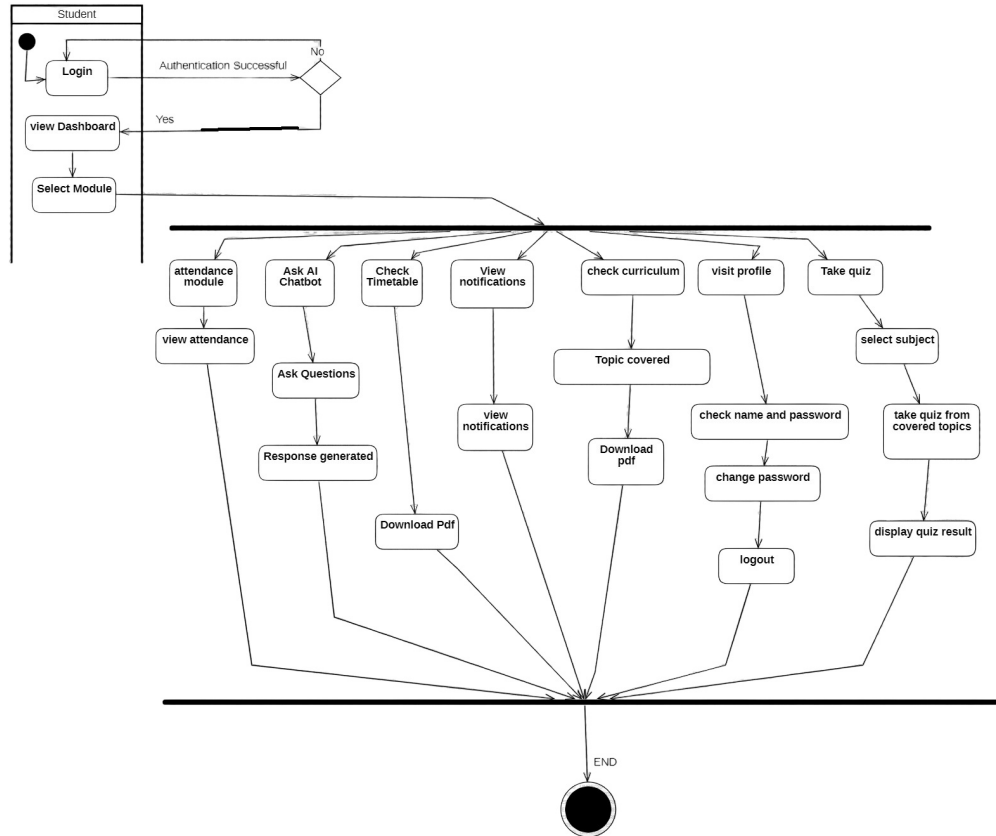


Figure 4.4 Student Activity Diagram

4.3.2 Use Case Diagram

The Use Case Diagram (Figure 4.5) provides a high-level view of how each user class interacts with the Smart Educational Companion. Three actors are represented: the **Student**, the **Teacher**, and the **Super Administrator**.

The **Teacher** is responsible for the operational management of academic content and student data. Teacher use cases include creating and managing quizzes, uploading course materials and attendance records, pushing notifications to students, viewing student performance reports, and interacting with the chatbot for timetable and curriculum queries.

The **Student** interacts with the learning and assessment features of the platform. Student use cases include logging in, viewing course materials, attempting quizzes, receiving AI-driven feedback, checking timetable and attendance, tracking achievements, and reading notifications.

The **Super Administrator** has a focused set of use cases limited to account management:

adding teacher accounts, removing teacher accounts, and viewing the registered teacher list. The super administrator does not share any use cases with the student or teacher actors. This separation of actors and use cases ensures that the functional boundaries of each role are clearly defined and that no role has implicit access to another role's features.

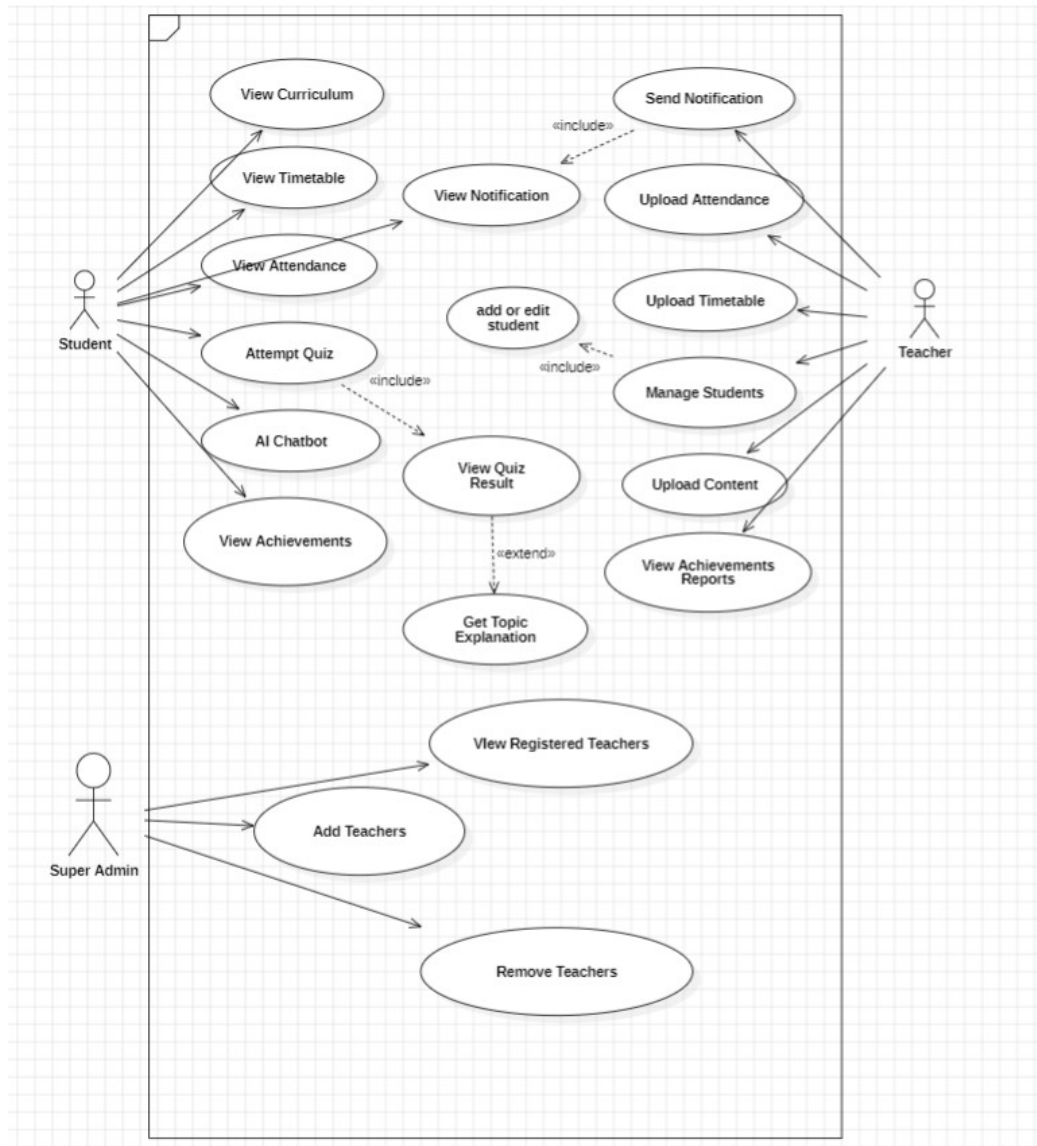


Figure 4.5 Use Case Diagram for Student, Teacher, and Super Administrator

4.3.3 Sequence Diagrams

Sequence diagrams reveal how the components of the system respond to the actions performed by the users according to their time order. They make the flow of control within the modules clear and help to identify which module handles each step during a particular operation. There are three sequence diagrams in total: the teacher sequence diagram, the super administrator sequence diagram, and the student sequence diagram.

Teacher Sequence Diagram

The Teacher Sequence Diagram (Figure 4.6) illustrates the step-by-step interactions that occur between components of the system as triggered by the teacher in a single session. The process starts with authentication of the teacher via the login page, which requests for a check of credentials from the backend and receives a role-based token. With access from the teacher dashboard, the teacher can upload course content and attendance information to the database, create quizzes, notify students using the notification service, and get the performance data of the students in the application layer.

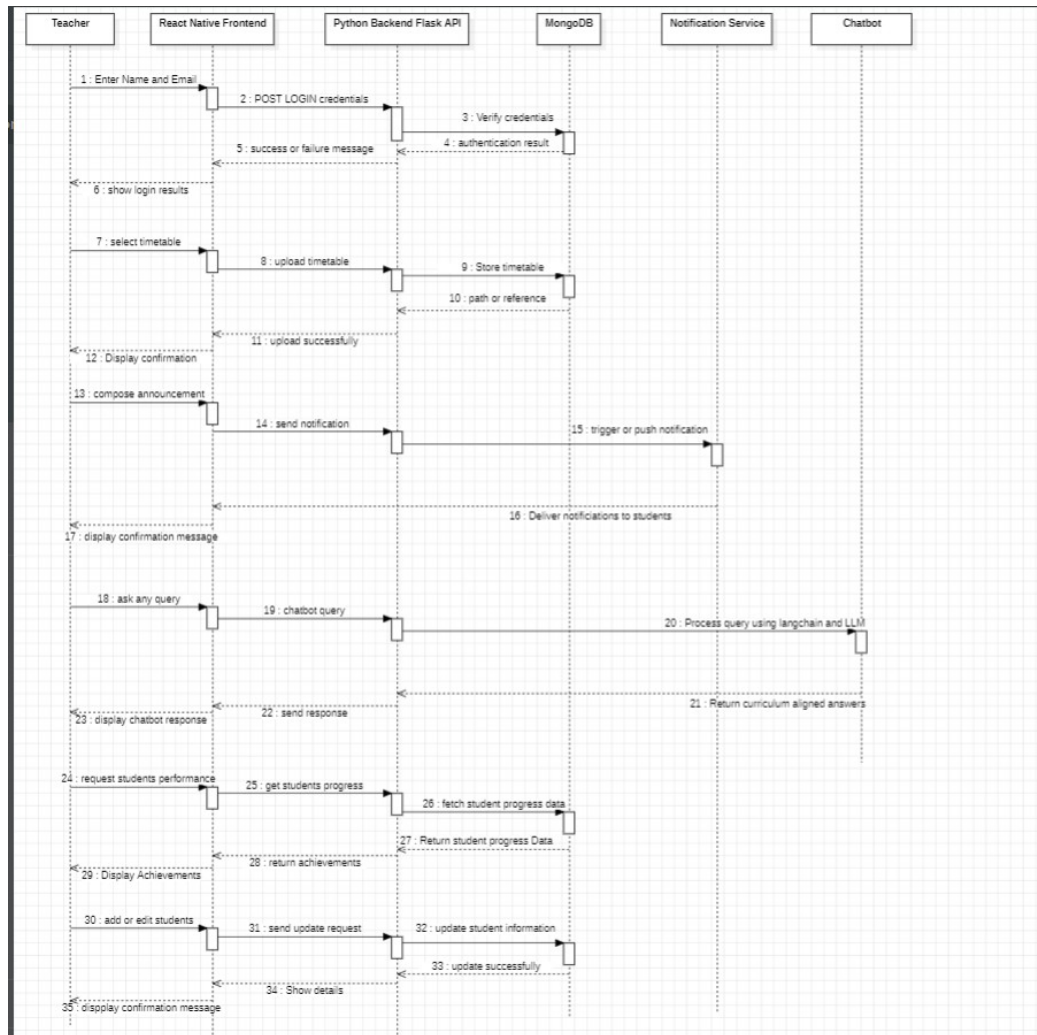


Figure 4.6 Sequence Diagram for Teacher

Super Administrator Sequence Diagram

Figure 4.7, the Super Administrator Sequence Diagram, presents the sequence of interactions for account management functions. The super administrator first logs into his or her account. After login, the super administrator can make a request for the list of the existing teachers, make a request for a new teacher's account creation, or make a request for deleting a teacher's account. The application layer will execute all those requests in turn.

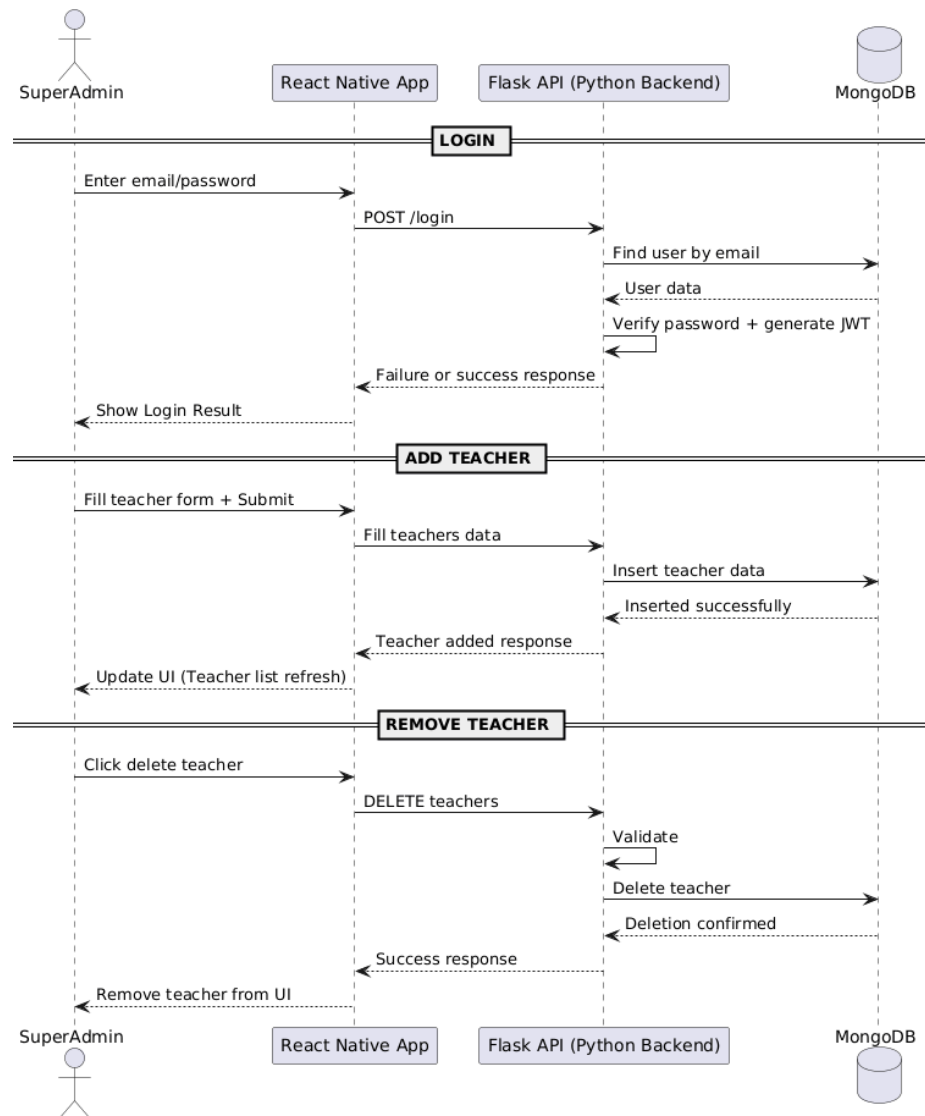


Figure 4.7 Sequence Diagram for Super Administrator

Student Sequence Diagram

The Student Sequence Diagram (Figure 4.8) depicts the sequence of operations a student performs. Upon successful authentication of the student role, the student is enabled to access course content from the content component and post answers for the quiz to the quiz engine that automatic evaluation and chatbot-generated feedback, query the chatbot directly for curriculum explanations, retrieve timetable and attendance data from the database, and check the notification screen for teacher announcements.

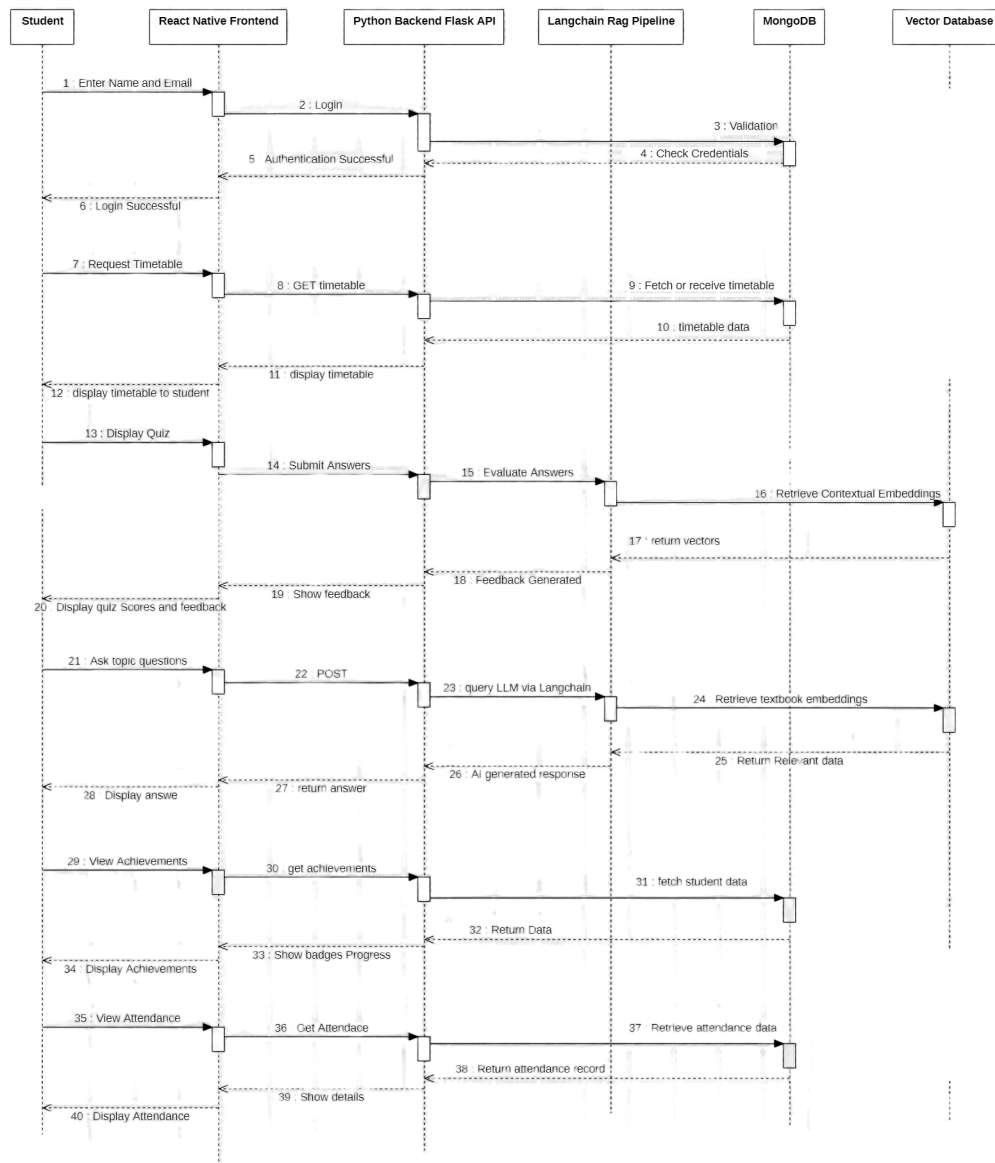


Figure 4.8 Sequence Diagram for Student

4.3.4 State Transition Diagrams

State transition diagrams map out every state a user can be in within the system and the conditions that move them from one state to the next. For the Smart Educational Companion, these diagrams cover all three user roles students, teachers, and the super administrator from the moment they log in through every module they can access until they log out.

Teacher State Transition Diagram

The Teacher State Transition Diagram (Figure 4.9) shows how a teacher moves through the system after authentication. From the teacher dashboard, the teacher can transition into any of the management states uploading course content, uploading attendance, sending notifications, or viewing student performance reports. Each state returns to the dashboard upon completion, and the session ends at logout.

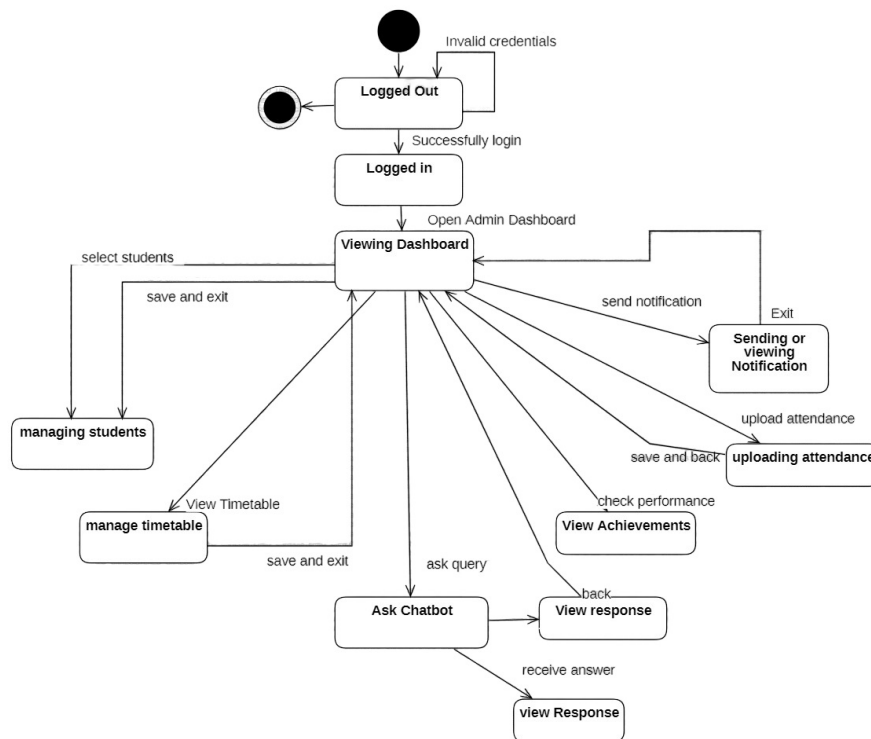


Figure 4.9 Teacher State Transition Diagram

Super Administrator State Transition Diagram

The Super Administrator State Transition Diagram (Figure 4.10) covers the narrow but critical set of states available to the super administrator. After completing OTP-verified login, the super administrator lands on the super admin dashboard, from which three state transitions are possible: viewing the registered teacher list, adding a new teacher account, or removing an existing teacher account. Each action returns to the dashboard upon completion. There are no transitions from the super administrator's session into any student-facing or teacher-facing states, and the session ends cleanly at logout.

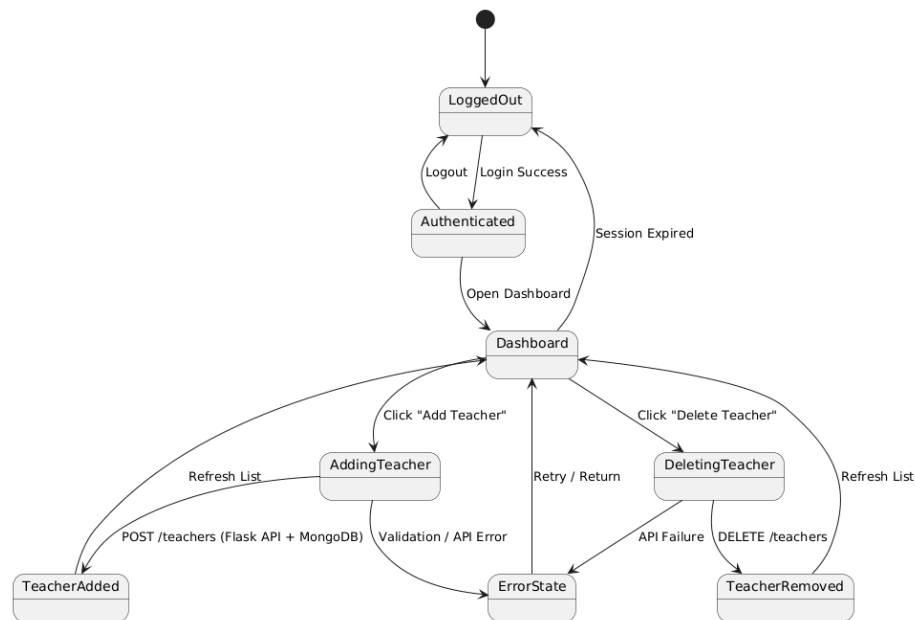


Figure 4.10 Super Administrator State Transition Diagram

Student State Transition Diagram

The Student State Transition Diagram (Figure 4.11) maps the full set of states a student can occupy during a session. After login, the student enters the dashboard state, from which they can transition into course material viewing, quiz attempt, chatbot interaction, timetable viewing, attendance checking, achievement tracking, and notification reading. Each module state returns the student to the dashboard once the activity is complete. The quiz attempt state has a direct transition into the result and feedback state, where AI-generated explanations are displayed before the student returns to the dashboard. The session ends at logout.

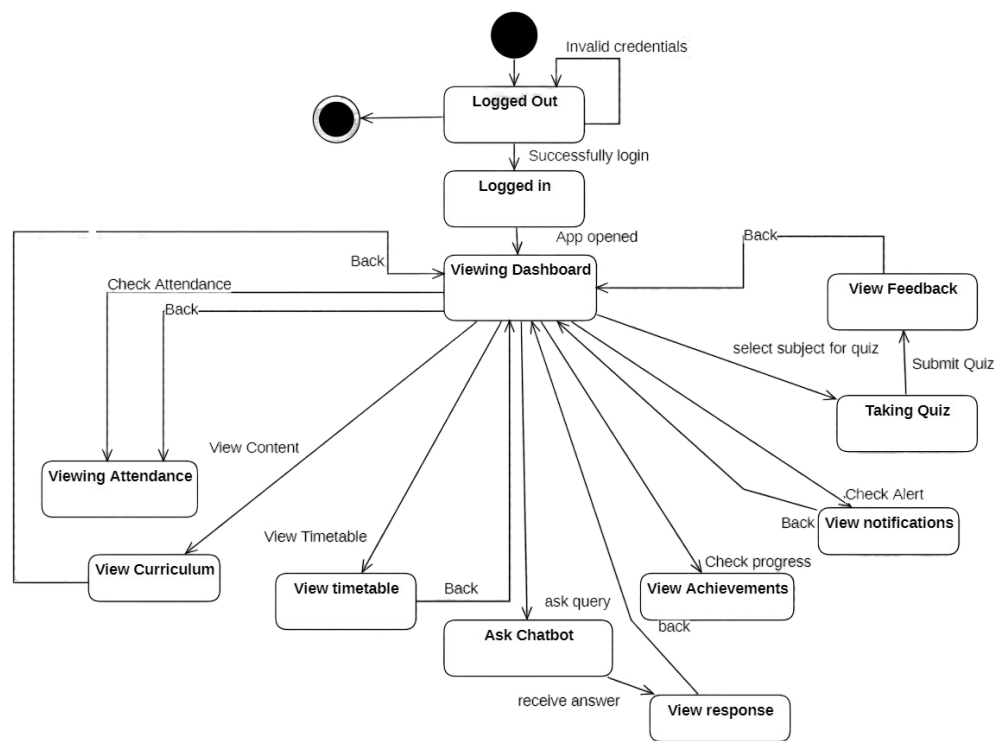


Figure 4.11 Student State Transition Diagram

System Class Diagram

The System Class Diagram (Figure 4.12) presents the static structure of the Smart Educational Companion, showing the major classes, their attributes and methods, and the relationships between them. The diagram covers all core modules Timetable, Curriculum, Quiz, Chatbot, Notification, Attendance, and Achievement and illustrates how the three role-based controllers interact with the underlying service classes. The Student class connects to the Quiz, ChatbotLog, Achievement, and AttendanceRecord classes through association relationships. The Teacher class connects to the Timetable, CourseMaterial, Notification, and AttendanceRecord classes. The SuperAdministrator class is associated only with the managing teachers

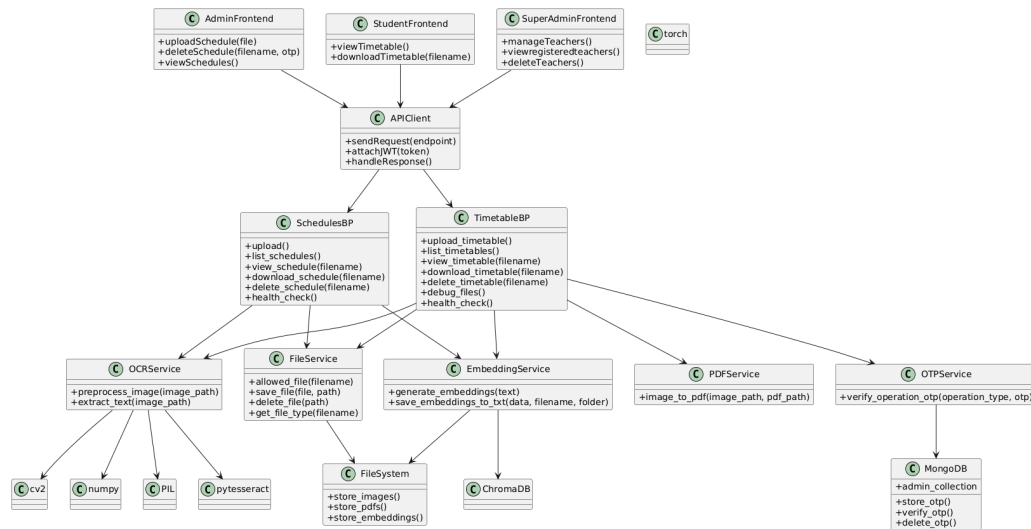


Figure 4.12 System Class Diagram

4.4 Data Design

The Smart Educational Companion handles data across several distinct modules: student records, attendance, quizzes, course materials, chatbot interactions, timetables, notifications, achievements, and account management for both teachers and the super administrator. These modules do not all store data in the same shape, so a database that enforces a rigid fixed schema would introduce unnecessary friction. For this reason, the system uses **MongoDB**, a NoSQL database that stores records as JSON-like documents and accommodates the variation in structure that exists naturally across these entities.

As another example, the student record holds the badges and progress data, which the teacher record lacks. The quiz record holds nested objects with questions that look nothing like a timetable record object. MongoDB handles the differences elegantly, without resorting to any workarounds with sparse columns.

To implement AI functionalities, the application couples a **vector database** with MongoDB. Embedding the content of the curriculum into the vector database allows for retrieval during chatbot queries. This implementation is crucial to achieve the functionality of the **RAG (Retrieval Augmented Generation) pipeline** developed using **LangChain**. RAG retrieves curriculum passages from the vector database and uses them along with an open-source language model to generate the response, ensuring accuracy and contextual relevance based on the Punjab Board's curriculum and not the internet.

In summary, through MongoDB, the vector database, and the RAG pipeline, the application is capable of handling diverse data types, responding quickly in real time, and generating relevant information from the curriculum.

4.4.1 Data Dictionary

The table shown below represents some of the main data entities within the Smart Educational Companion system. Each of these entities maps onto a collection within the MongoDB database.

Table 4.1 Data Dictionary for Smart Educational Companion

Entity	Attribute	Data Type	Description
Student	studentID	Integer	Unique identifier for each student.
	name	String	Full name of the student.
	email	String	Registered email address used for login.
	password	String	Encrypted password for authentication.
	progress	Float	Overall quiz performance percentage.
	badges	Array[String]	Achievement badges earned by the student.
Teacher	teacherID	Integer	Unique identifier for each teacher account.
	name	String	Full name of the teacher.
	email	String	Registered email used for teacher login.
	password	String	Encrypted password for authentication.
Super Admin	superAdminID	Integer	Unique identifier for the super administrator account.

Entity	Attribute	Data Type	Description
	email	String	Registered email used for super admin login.
	password	String	Encrypted password for authentication.
Quiz	quizID	Integer	Unique identifier for each quiz.
	topic	String	Subject and topic the quiz covers.
	questions	Array[Object]	Question text, answer options, and correct answer index for each item.
	totalMarks	Integer	Maximum marks available for the quiz.
	createdBy	Integer	Teacher ID of the teacher who created the quiz.
QuizResult	resultID	Integer	Unique identifier for each quiz attempt.
	studentID	Integer	Foreign key referencing Student.
	quizID	Integer	Foreign key referencing Quiz.
	obtainedMarks	Integer	Marks scored by the student.
	dateAttempted	DateTime	Timestamp of the quiz attempt.
	feedback	String	AI-generated feedback for incorrect answers.

Entity	Attribute	Data Type	Description
ChatbotLog	messageID	Integer	Unique identifier for each chatbot interaction.
	studentID	Integer	Foreign key referencing the Student who sent the query.
	query	String	Text input submitted by the student.
	response	String	Response generated by the RAG pipeline.
	timestamp	DateTime	Date and time of the interaction.
Timetable	timetableID	Integer	Unique identifier for each uploaded timetable.
	uploadedBy	Integer	Teacher ID of the teacher who uploaded the file.
	filePath	String	Storage path or URL of the uploaded file.
	parsedData	JSON	Structured schedule data extracted from the uploaded document via OCR.
	uploadDate	Date	Date on which the timetable was uploaded.
Notification	notificationID	Integer	Unique identifier for each notification.
	title	String	Short heading of the notification.

Entity	Attribute	Data Type	Description
	message	String	Full content of the notification.
	sentBy	Integer	Teacher ID of the teacher who sent the notification.
	dateSent	DateTime	Timestamp of when the notification was dispatched.
Achievement	achievementID	Integer	Unique identifier for each awarded achievement.
	studentID	Integer	Foreign key referencing the Student who earned it.
	badgeName	String	Name of the badge or reward awarded.
	dateAwarded	Date	Date on which the badge was granted.
AttendanceRecord	recordID	Integer	Unique identifier for each attendance upload.
	uploadedBy	Integer	Teacher ID of the teacher who uploaded the record.
	filePath	String	Storage path or URL of the uploaded attendance file.
	classID	String	Identifier for the class the record applies to.

Entity	Attribute	Data Type	Description
	uploadDate	Date	Date on which the record was uploaded.
UserAuth	userID	Integer	Identifier linking to Student, Teacher, or Super Admin.
	email	String	Registered email address used at login.
	passwordHash	String	Hashed and salted password string.
	otpCode	Integer	One-time password issued during verification.
	otpExpiry	DateTime	Expiry timestamp after which the OTP is no longer valid.

4.5 Human Interface Design

The Smart Educational Companion is built around the principle that every user regardless of their role should be able to find what they need quickly and complete their tasks without confusion. The system interface is divided into three parts: one for the students, another for the teacher, and the third for the super administrator. Each of the above users gets their own interface designed for their needs, although a uniform color palette, typography, and layout strategy have been maintained throughout the interface. The design was created in the **Figma** app and realized using the **React Native** framework.

For each user separately:

- **Teacher:** School teacher Logs in to get access to the teacher’s panel, where he/she can upload the calendars and attendance record, create and share quizzes, upload course accessories, send notifications to the students, and also see performance summaries of his/her students. All actions involving changes or deletion of some infor-

mation will be accompanied by prompts, successes, and error messages.

- **Student:** The user logs in to access study materials, view the timetable and attendance, pass quizzes and get AI feedback about the results, check achievements, and communicate with an AI chatbot. Feedback messages come in the form of scores, chatbot messages, badges, and messages from the teacher.
- **Super Administrator:** After logging in with a one-time password, the super administrator can get access to his/her panel, where the super administrator can view the list of registered teachers, add new teacher accounts, and delete already existing ones. This screen was specially made minimal because of the limited tasks the super administrator performs.

4.5.1 Screen images

The images below demonstrate screens from the developed application.

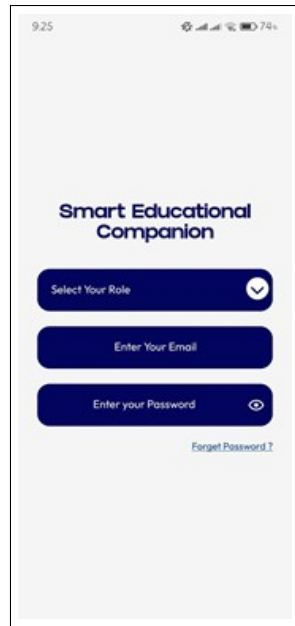


Figure 4.13 Login Screen

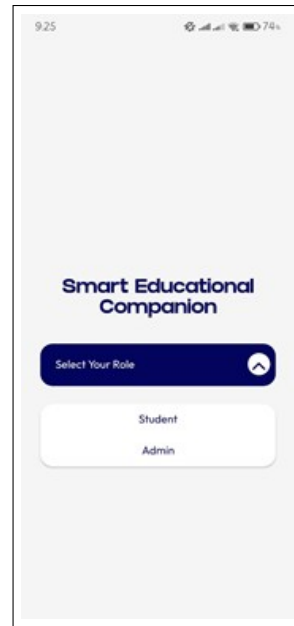


Figure 4.14 Role Selection Screen

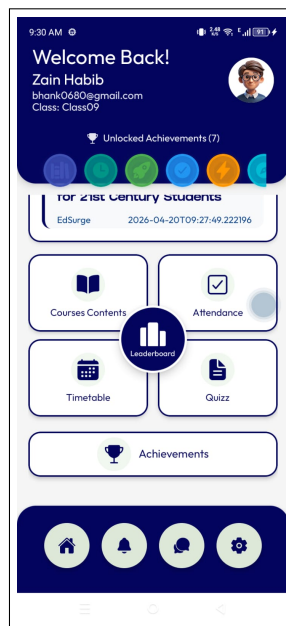


Figure 4.15 Student Dashboard

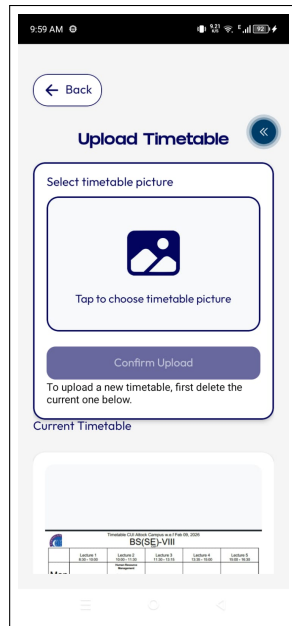


Figure 4.16 Timetable Screen

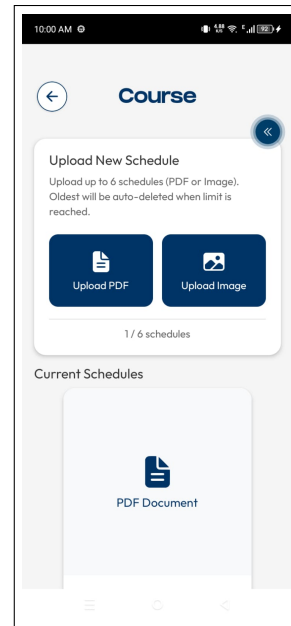


Figure 4.17 Curriculum Screen

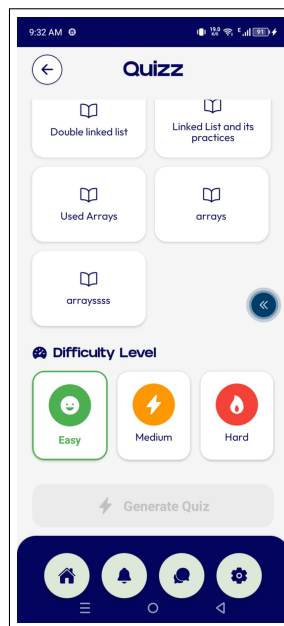


Figure 4.18 Quiz Screen

4.5.2 Screen Objects and Actions

All screens within the application contain interactive objects that relate to particular user actions. Below is the table listing objects presented on each key screen along with possible interactions:

- **Login Screen:**

- Objects: Field for email address, field for password, button for signing in, link for recovering password.
- Actions: Provide authentication details and press Sign In to proceed; click on the link to start the process of password restoration using OTP.

- **Screen for Choosing the User's Role:**

- Objects: Buttons or cards for choosing roles of Student, Teacher, and Super Administrator.
- Actions: Press on the card with the selected role to move to the next stage of the process; selection of the Super Administrator role requires OTP verification to ensure security.

- **Dashboard of the Student:**

- Objects: Menu with links to Courses, Timetable, Quizzes, Attendance, Chatbot, Achievements, and Notifications sections.
- Actions: Choose any module to go to it; navigate back to the dashboard from any module by pressing the back/home button.

- **Dashboard of the Teacher:**

- Objects: Tabs for Course Materials, Quizzes, Attendance, Timetable, Notifications, and Student Performance sections.
- Actions: Upload files, create quizzes, send notifications, and review performance statistics. Confirmation windows appear before any removal or publishing actions are taken.

- **Screen of the Super Administrator:**

- Objects: Display of a list of teachers, button for adding a new teacher, and button for removing a teacher from the database.
- Actions: Review all teachers who have registered on the website; click on the "Add" button to add a new teacher by entering their credentials; click on the "Remove" button to delete an existing teacher from the database.

- **Quiz Screen:**

- Objects: Question text, buttons with answers, button for submitting the quiz, progress bar.
- Actions: Choose an answer for each question, submit the quiz, see the result along with item-specific feedback with explanations from AI.

- **Chatbot Screen:**

- Objects: Text input field, Send button, scrollable chat window.
- Actions: Type an academic query and tap Send; the chatbot response appears in the chat window

4.6 Algorithms

This section presents the key algorithms that drive the Smart Educational Companion's core functionality. Each algorithm is described with its input, expected output, and step-by-step logic.

Algorithm 1: Login and Authentication

Input: User credentials (email, password)

Output: Role-specific dashboard or error message

Steps:

1. Show the login form with the fields for email and password.
2. Capture the input once the user clicks on the Login button.
3. Ensure that both fields have been filled out.

4. Authenticate credentials against the backend authentication API.
5. In case the credentials are authenticated, check the role from the session token received and route the user to the appropriate dashboard: Student Dashboard, Teacher Panel, or Super Administrator Panel.
6. In case of wrong credentials, show an error message without disclosing which input was incorrect.

Algorithm 2: Role Selection

Input: User role selection (Student, Teacher, Super Administrator)

Output: Interface corresponding to the selected role

Steps:

1. Show the three roles that can be selected on the role selection screen.
2. Wait until the user taps their desired role.
3. Mark the selected role to ensure the user understands the role they have chosen.
4. Save the selected role locally.
5. In case the Super Administrator role is selected, initiate the OTP verification process before continuing.
6. Upon successful verification or if the role selected is Student or Teacher, direct the user to the next screen.

Algorithm 3: Chatbot Interaction

Input: Student query text

Output: AI-generated curriculum-aligned response

Steps:

- Step 1. Get the input text from the student's chat area.
- Step 2. Present the message right away in the student's chat box.
- Step 3. Send the question to the RAG pipeline backend through API.
- Step 4. The pipeline creates an embedding for the question, finds related curriculum content from the vector DB, and sends it to the model.

Step 5. Receive the output response from the backend.

Step 6. Show the output in the chat box.

Step 7. Auto-scroll to the last message.

Algorithm 4: Quiz Evaluation

Input: Selected quiz topic and student's answers

Output: Score, item-level feedback, and AI-generated explanations

Steps:

1. Pull data related to quizzes from the backend that includes questions and their options.
2. Display questions along with multiple choices.
3. Save the user's selected option to each question as they advance through the test.
4. Send all user inputs to the backend for scoring once they submit the test.
5. Compare submitted inputs with the actual answers to determine the test score.
6. Upon receiving an incorrect answer, use the chatbot module to provide an explanation of the topic.
7. Display final score and all explanations.

Algorithm 5: Teacher Account Management (Super Administrator)

Input: Teacher account details or teacher ID for removal

Output: Updated teacher list

Steps:

1. Super administrator logs in via OTP-verified authentication.
2. The system retrieves and displays the current list of registered teacher accounts.
3. To add a teacher: capture name, email, and initial password; submit to the backend; create the account in the database; display a success confirmation.
4. To remove a teacher: select the account from the list; display a confirmation prompt; on confirmation, send a deletion request to the backend; remove the account from the database and update the displayed list.

Algorithm 6: Notification Dispatch

Input: Notification data (title, message) composed by teacher

Output: Notification delivered to all registered students

Steps:

1. Teacher composes a notification title and message body from the teacher panel.
2. On submission, the notification is sent to the backend notification service.
3. The backend stores the notification in the database and dispatches it to all registered student accounts.
4. Students receive the notification in real time within the app's notification screen.
5. If the device permits, a native push alert is also displayed on the student's phone.

4.7 Tools and Technologies

Table 4.2 Technical Stack with Implementation Rationale

Tool / Technology	Rationale
Figma	Wireframing and prototyping tool used to design all three role-based interfaces before development began, catching layout problems early.
Visual Studio Code	Code editor for both the React Native frontend and Python backend, with extensions supporting JavaScript, TypeScript, and Python in a single environment.
Android Studio	Provides virtual Android devices for testing the app across different screen sizes and API levels without needing multiple physical phones.
Expo	Used for rapid prototyping during early development. Live QR preview on physical devices sped up the UI testing cycle.
React Native	Core mobile framework producing a single JavaScript codebase that compiles to native Android components. Expo simplified navigation and build setup.
Python	Backend language handling API requests, business logic, AI orchestration, and database communication. Strong ML library support suited the chatbot requirements.
FastAPI / Flask	Python web frameworks exposing REST APIs to the mobile frontend. FastAPI for validation and docs, Flask for simpler routes.

Tool / Technology	Rationale
MongoDB	NoSQL database storing student records, teacher accounts, quizzes, attendance, timetables, notifications, and chatbot logs. Flexible schema handles structural variation across entities.
Vector Database (ChromaDB / FAISS)	Stores curriculum content embeddings. Performs semantic search during chatbot queries to retrieve relevant Punjab Board passages before response generation.
LangChain	Orchestrates the RAG pipeline: embedding, retrieval, context assembly, and model invocation. Modular design allows component swapping without pipeline restructuring.
RAG Pipeline	Grounds chatbot responses in verified Punjab Board content instead of generic training data, preventing plausible but incorrect answers.
Hugging Face (Gemma, LLaMA)	Open-source language models generating chatbot and quiz feedback text. Self-hosted deployment avoids ongoing per-query API costs.
Git + GitHub	Version control and collaboration platform. Git managed feature branches; GitHub handled code review and merge tracking.

4.8 User Interface

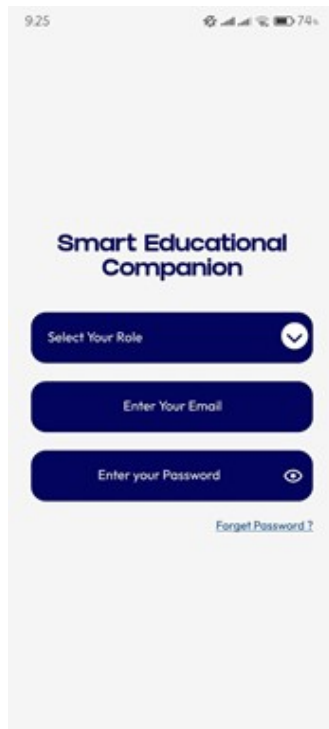


Figure 4.19 Login Screen

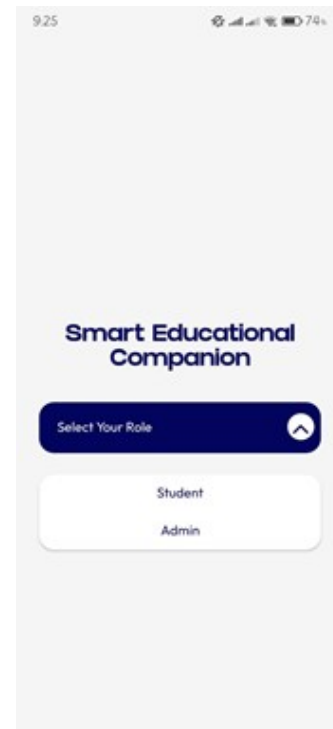


Figure 4.20 Role Selection Screen

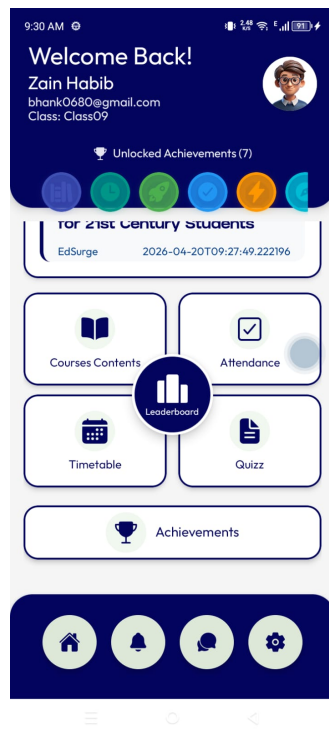


Figure 4.21 Student Dashboard

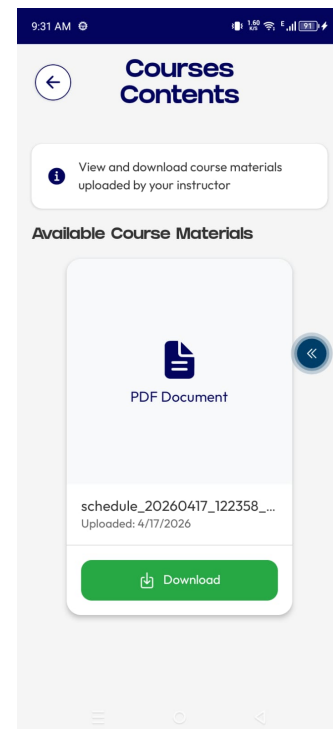


Figure 4.22 Curriculum Screen

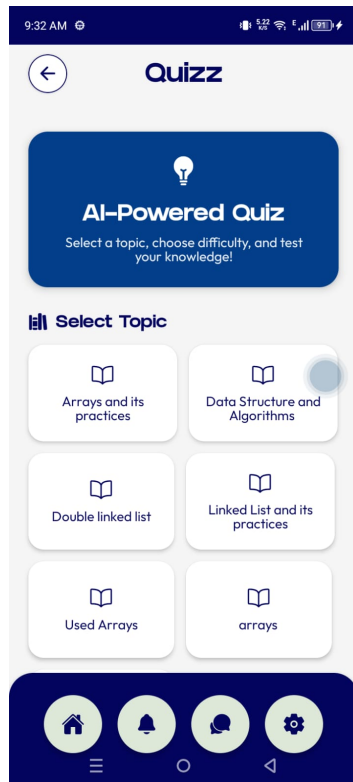


Figure 4.23 Quiz Screen

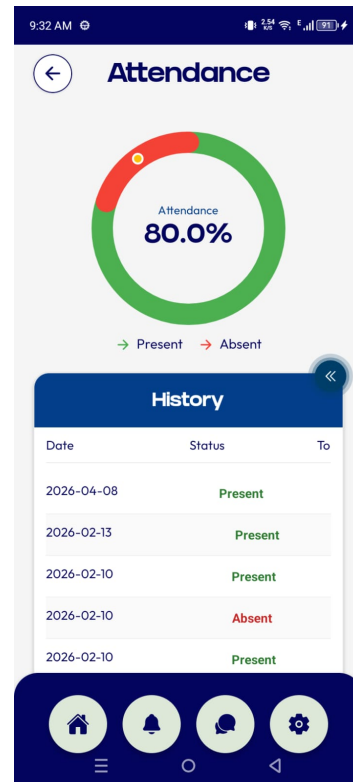


Figure 4.24 Attendance Screen

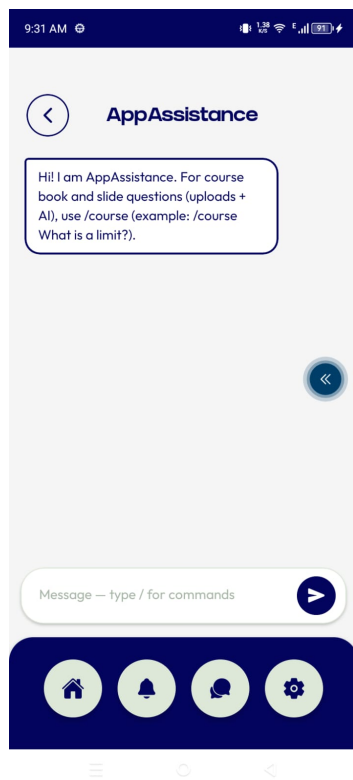


Figure 4.25 Chatbot Screen

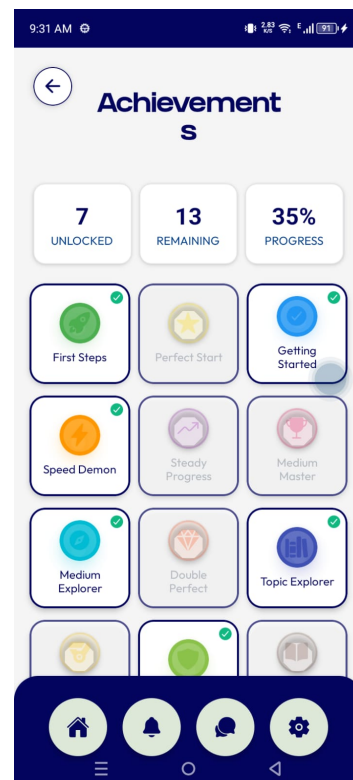


Figure 4.26 Achievements Screen

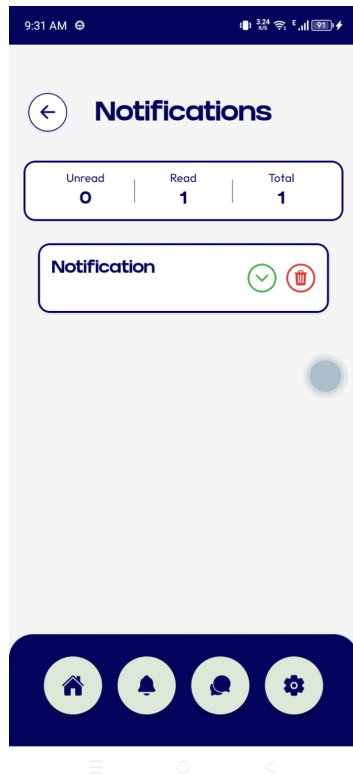


Figure 4.27 Notification Screen

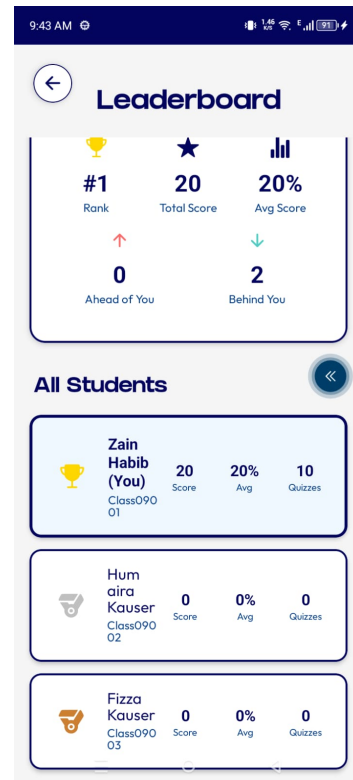


Figure 4.28 Leader Board Screen

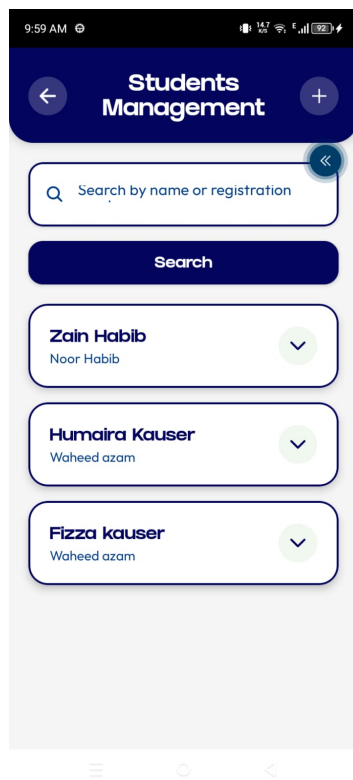


Figure 4.29 Students Management Screen

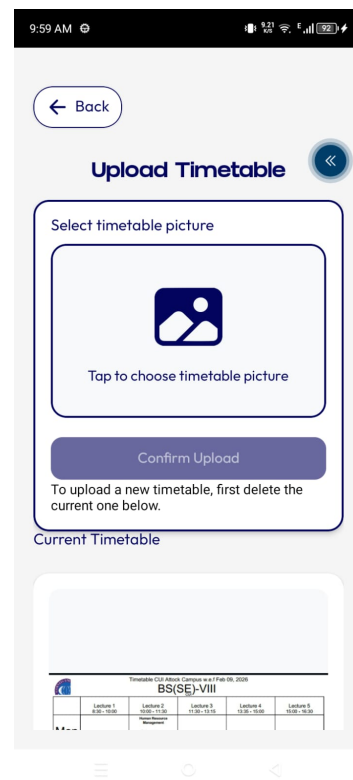


Figure 4.30 Upload Schedule Screen

4.9 Chapter Summary

This chapter presented the overall design and architecture of the Smart Educational Companion system. It detailed how different modules and subsystems—such as the mobile application, backend services, database, and AI components—interact to deliver a seamless learning experience. The architecture of the system was depicted through an illustration showing the data flow among the components, with particular focus on modularity, scalability, and maintainability. In addition to that, the chapter provided an overview of the user interface design in terms of layout, screen features, and interaction processes that the user goes through when using the software. In addition, this part included information about the screen objects and actions that users could take advantage of when interacting with the software. Lastly, the tools and technologies subchapter presented the software stack that was selected to develop the system.

Chapter 5

Testing and Evaluation

5.1 Chapter Overview

This chapter details the testing process used in the Smart Educational Companion to assure that each and every module functions properly within the conditions it is designed to operate in. Tests were performed in several stages, including unit tests to determine if separate components work correctly, functional tests to check if the processes specified by the user work end-to-end, business rules validation to ensure that business logic is correctly enforced, and integration tests to ascertain that all the modules interact properly when combined into one system. The results described in this section prove that the system works properly in all tested situations.

5.2 Testing Introduction

It cannot be said that any piece of software can be considered complete until it has passed through systematic testing. Testing is the step that tells whether implementation meets requirements, how well it copes with edge cases, and whether all separate modules, which can be developed separately, interact correctly when put together. Testing of the Smart Educational Companion was started on component level and then moved up to cover user journey testing, business logic validation, and even integration between different modules. Not only does it check if functionality works when invoked properly, but also how the system reacts to wrong or stressed conditions.

5.3 Unit Testing

Unit testing is the process where each individual module is checked separately to see whether it functions according to the requirements. Module is considered to pass unit testing if for any correct input it provides expected output and correctly processes wrong inputs without crashing and failing silently.

5.3.1 Unit Test Cases

Table 5.1 shows results of validation of separate modules of the Smart Educational Companion on a unit level.

Table 5.1 Unit Test Cases

Module Under Test	Test Description and Expected Behavior	Result
Login Module	Successfully authenticated all three users. If the credentials of the student are correct, they should land on the student dashboard, while the teacher should land on their panel, and the super administrator on theirs after the successful verification of OTP. In case of wrong credentials, the process should fail with an appropriate message without disclosing any of the fields.	Pass
Curriculum Module	Tested whether the module shows topics related to Punjab Board, timetables on a weekly basis, and materials related to various subjects. Also verified whether chatbot answers the query using this information.	Pass
Quiz Module	Workflow for test-taking process from displaying questions to submission of answers and scoring was tested. It was checked that scoring system correctly determines correct and incorrect answers, AI-assigned feedback is generated for each incorrect answer, and points and badges are given accordingly.	Pass
Chatbot Module	Raised various curriculum-based questions and confirmed that answers are based on Punjab Board material available through RAG pipeline. Conducted tests on academic student queries as well as time-table queries to ensure that the bot is capable of answering both types.	Pass
Timetable & Attendance Module	Tested the teacher upload workflow for timetable and attendance files, including OCR-based data extraction from uploaded images. Verified that extracted data is correctly stored and displayed to students on their dashboard.	Pass

Module Under Test	Test Description and Expected Behavior	Result
Achievement Module	Verified that quiz performance correctly updates student achievement points and triggers badge awards when thresholds are met. Confirmed that teachers can view aggregated achievement summaries for their students.	Pass

5.4 Functional Testing

Functional testing validates that the system delivers the end-to-end workflows it was built to support. Unlike unit testing, which isolates individual components, functional testing exercises complete user journeys from start to finish, confirming that all the parts work together correctly to produce the intended result. A feature passes functional testing when a real user action produces the expected outcome without errors, delays, or incorrect behavior.

5.4.1 Functional Test Cases

Table 5.2 presents the functional validation results for the major workflows in the Smart Educational Companion. Each test represents a complete user action from initiation through to completion.

Table 5.2 Functional Test Cases

Workflow Tested	Test Description and Expected Behavior	Result
Multi-Role Navigation	Verified that logging in with student credentials directs to the student dashboard with access to course materials, quizzes, timetable, attendance, chatbot, and achievements. Verified that logging in with teacher credentials directs to the teacher panel with access to content uploads, quiz creation, attendance management, and notifications. Verified that logging in with super administrator credentials (after OTP verification) directs to the super admin panel with access only to teacher account management. Each role must have no access to the other roles' interfaces.	Pass
Quiz Workflow	Tested the complete quiz flow from student selection through question display, answer submission, automatic scoring, and AI-generated feedback delivery. Verified that quizzes load within 15 seconds, scoring completes immediately after submission, feedback explanations appear for every incorrect answer, and achievement points and badges are updated automatically based on the result.	Pass
Chatbot Interaction	Tested the student chatbot workflow from query submission through RAG pipeline processing to response display. Verified that curriculum-related queries return accurate, Punjab Board-aligned responses within 15 seconds, and that the chat history is preserved during the session.	Pass

Workflow Tested	Test Description and Expected Behavior	Result
Notification Delivery	Tested the teacher notification workflow from composition through backend dispatch to student delivery. Verified that notifications composed by a teacher appear on all registered students' notification screens within 10 seconds and, where device settings allow, trigger native push alerts on students' phones.	Pass
Timetable & Attendance Updates	Tested the teacher upload workflow for timetables and attendance records. Checked that upon upload of the timetable/attendance file by a teacher, the new data gets reflected instantly on the relevant students' dashboards without any need for manual refresh.	Pass
Achievement Tracking	Also tested the entire achievement calculation workflow, upon completion of a quiz by a student. Checked that upon quiz completion, the total points are calculated again, badges are granted if new levels have been achieved, and the new achievements appear on both the student dashboard and teacher performance screen.	Pass

5.5 Integration Testing

Integration testing is aimed at checking that individual modules, which were tested in isolation, will work as intended within the entire system. Integration testing will check that data will flow across module borders smoothly, modules will call each other as intended, and no undesirable issues arise during their interactions.

5.5.1 Test Cases for Integration Testing

Test cases for integration testing are provided in Table 5.4 below. The results of the integration validation of essential interactions among cross-module processes of the Smart Educational Companion system are listed in Table 5.4 below.

Table 5.3 Integration Test Cases

Modules Integrated	Test Description and Expected Behavior	Result
Login Module + Backend Authentication	Performed tests for validating credentials on the back end, for returning a valid session token containing valid role information, and for redirecting the frontend to the appropriate dashboard, whether it is the student dashboard, teacher dashboard, or super admin dashboard, according to the role information returned in the session token.	Pass
Quiz Module + Achievement Module	Conducted testing whether the result from a student completing the quiz is successfully passed by the Quiz Module to the Achievement Module for calculating the points and awarding new badges, along with reflecting the results on the students' and teachers' dashboards.	Pass
Chatbot Module + Curriculum Module	Checked that the Chatbot Module accurately fetches the Punjab Board's curriculum information using the RAG pipeline from the Curriculum Module for a particular student query and ensures that the response includes information on the right subject and topic only.	Pass
Notification Module + Timetable Module	Validated whether uploading of the new timetable by the teacher in the Timetable Module creates an auto notification from the Notification Module to all registered students that the timetable has been updated.	Pass
Timetable Module + Attendance Module	Confirmed that the attendance data entered into the system via the Attendance Module would be properly stored and linked to the timetable entry. Thus, the student can access his schedule and attendance details simultaneously on his dashboard.	Pass

Modules Integrated	Test Description and Expected Behavior	Result
Full System Integration	Performed an end-to-end test for an entire process flow that starts when the user logs in, checks their timetable, takes a quiz, gets AI feedback on the quiz, checks their attendance, achievements, and teacher notification. The process continues when the teacher logs in, uploads course content, creates a quiz, sends notifications, and analyzes the performance of their students. Super administrator logs in and creates a teacher's profile. Made sure that there are no conflicts in data or module interactions, and that every user cannot perform operations reserved for other roles.	Pass

5.6 Chapter Summary

This chapter presented a detailed account of the entire testing procedure performed on the Smart Educational Companion prior to its implementation. Testing was performed at four levels: unit testing to ensure that each module functioned as expected in isolation; functional testing to ensure that the overall workflow of the application resulted in the desired outcome; business rule testing to ensure that each role had access only to specific features without violating any constraints; and finally, integration testing to ensure that separate modules integrated seamlessly into the system.

The test cases at all four levels successfully passed, thereby proving that the system functions flawlessly under normal operations, handles incorrect data input without failing, restricts access based on roles without leakage, and ensures data integrity across all modules. This clearly indicates that the Smart Educational Companion satisfies the requirements mentioned in Chapter 3.

Chapter 6

Conclusion and Future Work

6.1 Chapter Overview

With this chapter, we conclude the document with a summary of the accomplishments of the Smart Educational Companion, an overview of the results of this work, and an analysis of the possible areas for further development of the platform.

6.2 Conclusion

The Smart Educational Companion was conceived to tackle specific issues in Pakistani secondary education, including students that had no means of accessing academic help after regular classes, excessive workload of teachers in performing routine academic management, and lack of institutional mechanisms of managing the operational aspect of the curriculum organization process. The platform resolves these issues using artificial intelligence, automated processes of providing assistance, and well-separated roles of users and managers, ensuring that every part of the educational experience can be managed in a highly customized way.

The main achievements of the project are listed below:

- **Three-Role Architecture:** The platform is implemented using three different user roles, including students, teachers, and a super administrator. Every role is limited to performing certain actions and accessing only relevant data, which helps keep the platform organized, restricts unauthorized access, and isolates management tasks from daily educational operations.
- **Artificial Intelligence-Based Assistance:** The platform uses a curriculum-oriented chatbot to provide students with accurate and timely responses that refer to the knowledge base of the Punjab Board instead of generic online sources. It provides detailed explanations to all questions asked, and automatically generates feedback on wrong quiz answers, not requiring teachers to manually check every response.
- **Automatic Academic Management:** Teachers only need to upload timetable, attendance sheets, and course material once, as the system will automatically distribute them to every student in real-time without the need for any additional actions. Automatic notifications will be sent as soon as a change happens, reducing the risks of

misunderstandings.

- **Gamification of Progress Monitoring:** Points and badges for completing quizzes are automatically assigned to students, allowing them to track their progress and teachers to identify which students need special attention.
- **Mobility-Focused Design:** All services of the platform are available to students and teachers using Android devices, which are widely spread throughout Pakistan. The design was made so that every feature can be reached within two taps from the main dashboard page.

The platform highlights that AI and mobile technology can be utilized in Pakistani secondary education as long as they are developed considering the specifics of the local environment. In particular, the platform operates efficiently taking into account budgetary limitations, varying degrees of the device's quality, and the peculiarities of the curriculum structure as compared to foreign platforms and leads to notable improvements in providing students with necessary information and assistance, increasing teacher efficiency, and improving management control.

6.3 Future Work

The Smart Educational Companion is aimed at fulfilling the main goals of the project. However, the platform can be further developed by expanding its functionality and capabilities for addressing the following issues:

- **Broader Scope of Subjects and Grade Levels:** Currently, the platform focuses on the subject matter for Class 9 students under the curriculum of the Punjab Board (Physics, Chemistry, Computer Science). Further, the platform may incorporate new subjects such as Mathematics, Biology, English, cover higher grades (Classes 10-12), and address other provincial curriculums (Sindh, KPK, Balochistan).
- **Handling Queries Out of Syllabus:** While the chatbot provides students with detailed answers related to the curriculum of the Punjab Board, any query outside the board curriculum is denied. Therefore, a mixed approach can be used where the bot

will try to answer any educational question, indicating whether the response is approved by the board.

- **Incorporating Predictive Analytics:** Currently, the platform tracks students' progress in a reactive manner. The inclusion of predictive analytics would allow the platform to identify the students that are likely to have problems in time, providing teachers with opportunities to prevent potential issues.
- **iOS and Web Platform:** The application operates under the Android OS. The extension to the iOS would provide students with more options when it comes to using the application. Also, the creation of a web-version of the platform would enable teachers and super-administrators to manage the system via the computer.
- **Offline Mode:** At the moment, the platform requires an Internet connection for its work. Offline mode for core learning functions (course materials and quizzes, previous quizzes of students, cached chatbot answers) would make the application accessible even during Internet outage.
- **Voice Input and Accessibility Features:** Including voice input as a chatbot query input method will enable learners having difficulties typing and those who prefer verbal communication. Incorporating screen reader compatibility and dynamic text size options will enhance accessibility features for visually impaired students.
- **Improved Gamification:** Currently, badges and points awarded in the achievement system are on an individual level. It is suggested that class-wide leaderboards, weekly challenges, and team-based objectives will encourage friendly competition without undermining the learning experience.
- **Parental Interface:** Although parents are not considered among users at this stage, implementing a read-only parental interface will provide guardians with real-time information on their child's progress in quizzes, attendance status, and achievements without logging in to their account.

6.4 Summary of Chapter

In summary, this chapter marked the final part of the document, reflecting on the results of the development of the Smart Educational Companion and suggesting possible paths of improvement. The conclusions section highlighted the following benefits provided by the educational platform under discussion: three clearly defined user roles; AI-based chatbot trained on content from the curriculum of Punjab Board instead of any common knowledge base; automated time table, attendance, and notification management eliminating manual distribution effort; gamified achievement system making student progress visible; and mobile-first design allowing users to access the application on devices commonly used by learners and teachers in Pakistan.

Directions for future research and development were provided in the form of eight potential extensions of the current application including expanded curriculum coverage, mixed handling of out-of-curriculum queries, predictive analytics to detect academic issues early, cross-platform iOS/web compatibility, offline access option, voice input and improved accessibility, enhanced gamification elements, and the introduction of a parental interface.

Overall, this chapter shows that the developed educational platform successfully solves the formulated problem while remaining easy to expand in the future.

Chapter 7

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